

Home Lab workshop

Private DNS, Centralized Log, Basic Monitoring Stack
- Jeremy, Jun Hong

Jeremy

- Previously Malware Analyst for 5 years
- Currently Senior Cyber Security Engineer at Recorded Future
- My motivations for homelab:
 - Completed GNFA → how to hone skills?
Answer: Homelab
 - Malware Analyst → how to test workflow?
What automated pipelines are out there?
Homegrown malware intelligence? Answer:
Homelab
 - Family → Setting up services for family use
- Not an expert



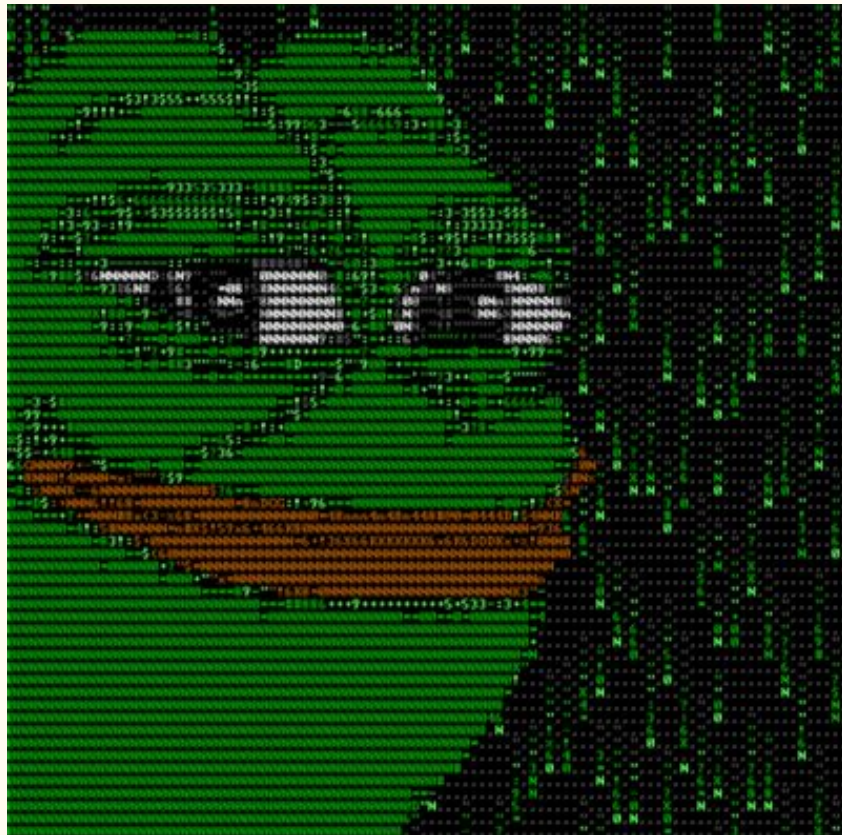
Jun Hong

- Generalist in Cyber Security
 - DFIR
 - SecOps
 - Cloud Security & Policy
- Currently Cyber Security Engineer at ICE
- My motivations for homelab:
 - Started out with blocking ads throughout home network
 - Went down rabbit hole of replacing every subscription and SaaS tools with self-hosted open source options
 - Learn and apply IT concepts (e.g. networking, logging & monitoring) on a smaller scale before applying in enterprise environments at work
- Not an expert too



Expected background

- New/interested in Cyber Security
- Some basic tech familiarity expected:
 - Have used VMware before
 - Used Linux before
 - Can read YAML or similar data serialization format (i.e., JSON)
 - Heard of **docker**
 - Basic IP + DNS knowledge



Pre-prep before workshop

- ❑ Hardware requirements:
 - ❑ Additional 64GB storage
 - ❑ At least 16 GB RAM
 - ❑ No ARM-based machines
- ❑ Install VMware Workstation Pro
 - ❑ This is free.
 - ❑ Watch this [Youtube video](#) for instructions
 - ❑ Read this [blogpost](#) to follow the guide
- ❑ Download and install Proxmox VE ISO in VMware
 - ❑ Official site [here](#) to download Proxmox VE 9.2 ISO Installer
 - ❑ Do not install on bare-metal for this workshop!
- ❑ Set up Proxmox in Network NAT Mode (do not forget your password lol)
- ❑ Use this [document](#) to guide you.



Workshop objectives

Goals:

- What is a homelab?
- Why is it useful?
 - For common folks
 - For cybersecurity professionals
- Getting started in homelab experience
 - Equipping you with know-hows to get started at home
 - What resources to help you continue developing your lab



Agenda



Part 1: Foundations 30 mins

- What is a Homelab?
- Why is it useful?
- What do I need to get started?
- (SIMPLE) Threat Model my homelab



Part 2: Hosting a DNS 45 mins

- What is Proxmox?
- Hosting a Docker management service on Proxmox
- Spinning up DNS server stack in Dockge



Part 3: Monitoring Logs/Metrics 45 mins

- What are logs and metrics?
- Why do we want to collect them?
- How to collect them?



Part 4: What next? 30 mins

- Exposing homelab beyond your home network
- What is Infra as Code?
- How to spin up components on Proxmox with IaC
- Brief touch and go concepts

Part 1: Foundations

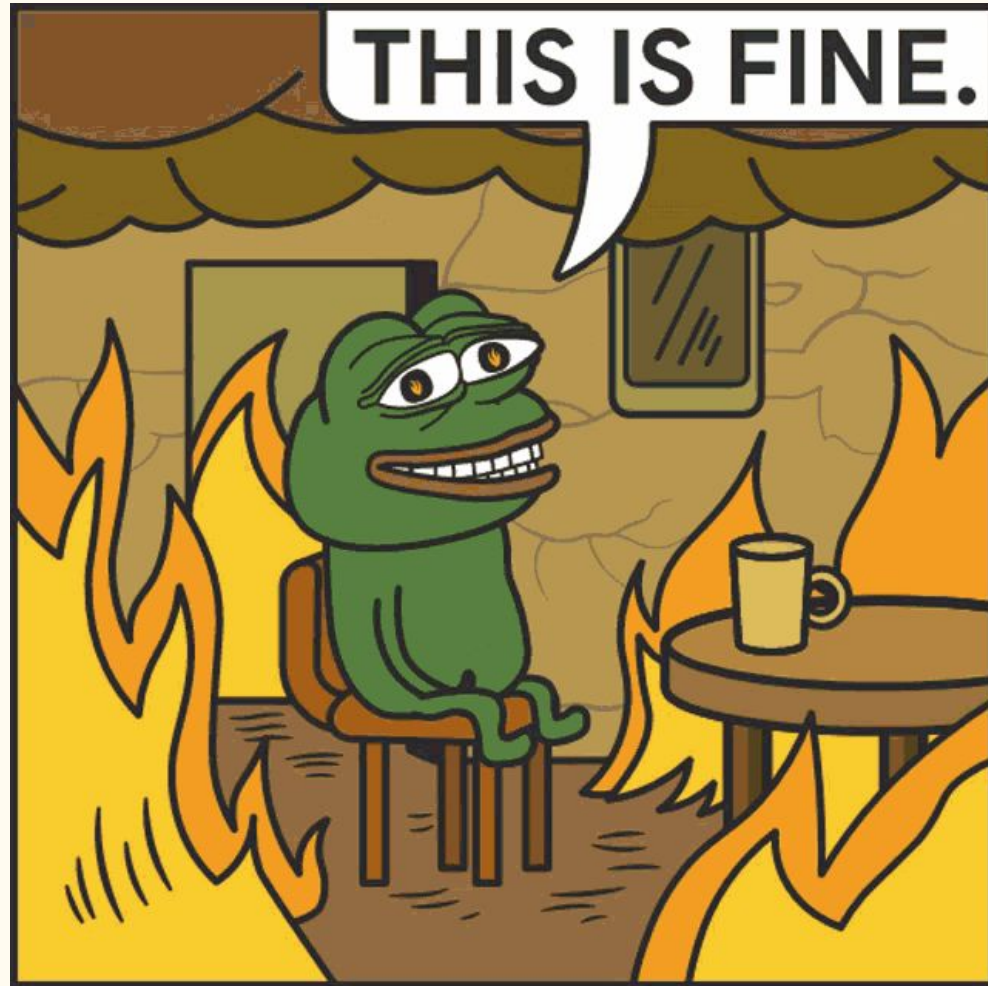


What is a homelab?

What it is: A place to build your enterprise IT skills—or just a great excuse to annoy your family by using the active home network as your personal danger zone.

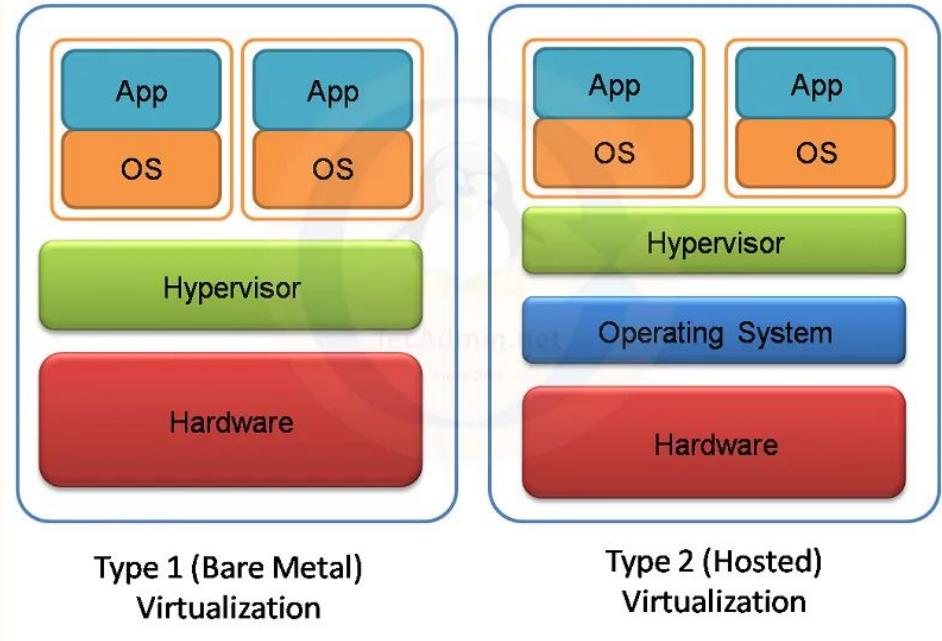
The modern approach: A virtual playground fully contained inside a single (or cluster) hypervisor server.

The internet exposure: Entirely up to you (and your risk tolerance). Discuss in Part 4



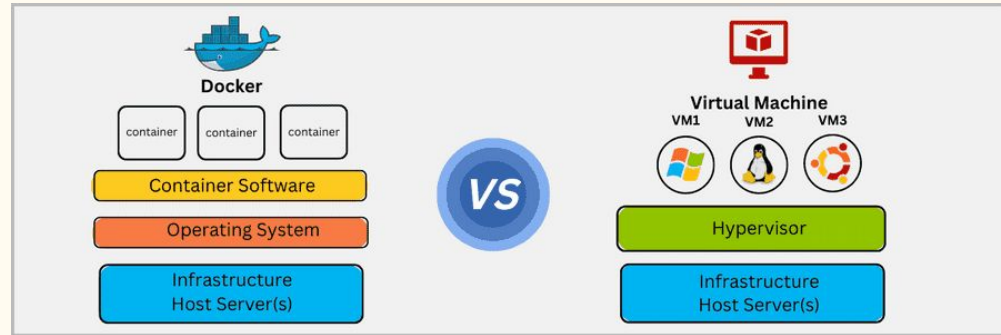
What is a Hypervisor?

- **The Allocator:** Pools actual CPU, RAM, and storage, hosting services in virtual environments.
- Type 1 virtualization is preferred in Homelab
 - Ease of creating new services
 - Much less bloat (Proxmox only uses 32-64GB of storage!)



What are containers vs Virtual Machines (VMs)?

- **OS-Level Isolation:** Each VM is its own OS. In Docker, each container shares the Host OS's kernel
- **Ultra-Lightweight:** Containers are extremely lightweight, used by software developers to ensure normalized environment for staging purposes



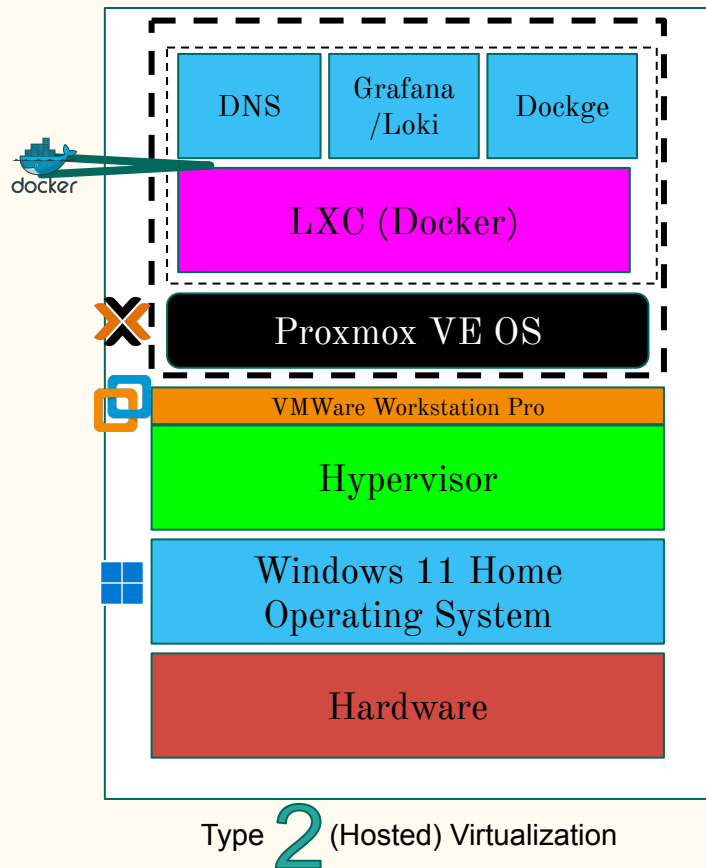
<https://diadem.in/blog/docker-vs-vm-hypervisor-vs-docker/>

Current setup:

- Proxmox in VM,
- containers in Proxmox

Our workshop architecture

- VMWare Workstation Pro is a software layer, acting as a Type 2 (Hosted) Virtualization
- If using Windows 11 Pro with Hyper-V installed, then your Windows acts as a Type 1 (Hybrid) Virtualization
- Beyond this workshop, but sharing for knowledge



- **The Ecosystem:** A self-hosted sandbox with active home network, and private data into one controlled environment.
- **The Core Node:** Physical hardware (like a mini PC) running a base OS to host your entire virtual workload.
- **Data Ownership:** Replacing big-tech cloud platforms with self-hosted alternatives for media, photos, and automation.

[illegible]

Addressing the elephant.

MAJOR DISCLAIMER

- **The Reality:** Platforms like Plex and Jellyfin are just the media players - they do not provide the content.
- **The Responsibility:** You are entirely responsible for sourcing, owning, and legally acquiring your own media library; we are sharing this because when you search about homelab, these topics will inadvertently be mentioned. The instructors and div0 are not responsible for the actions you choose to take after this workshop



Why do I need a homelab?

- **Ditch the Subscriptions:** Hate paying monthly fees to Apple, Google, Amazon, or Microsoft? Build a private cloud and pay the power bill instead!
- **Ultimate Privacy:** Don't trust what BIG CORPORATIONS are doing with your data? Host it yourself. Your files, your eyes only – privacy, baby!
- **Why pay?:** almost all services that you want to get but unwilling to pay are 90% likely to have an open-source alternatives



Subscription vs Owning (Cost Diff)

Cost Component	Homelab (e.g., 16 TB Usable Setup)	Subscribing to Services (Cloud Equivalent)
<i>Initial Setup (CapEx)</i>	\$1000–\$1,500 (Server hardware + 5x 4TB WD Red Plus HDDs @ \$212/pc, running at RAIDz1)	\$0
<i>Monthly Storage Cost</i>	\$0 (Incremental cost for adding data)	\$90/month (for only 3 users): Dropbox \$1080/year
<i>Monthly Compute Cost</i>	\$0 (Host dozens of Docker containers/VMs for free)	\$40–\$100/month (To run equivalent always-on VMs in AWS/GCP) \$480-\$1200/year
<i>Monthly Power/Utility</i>	\$5–\$25/month \$60-\$300/year	\$0
<i>Data Transfer/Egress</i>	\$0 (Limited only by your home internet plan)	\$90 per TB downloaded from enterprise object storage
<i>Yearly estimate cost</i>	(1st year): \$1500 (one-time cost) + (recurring): \$300 (monthly power/utility bill)	\$1080 (Dropbox subscription) + \$1200 (Estimated AWS services) = \$2280
<i>5 Yearly estimate cost</i>	\$300 * 5 (power/utility) + \$1000 (replacement of HDDs) = \$2500	\$2280 * 5 (yearly subscription costs) = \$11400

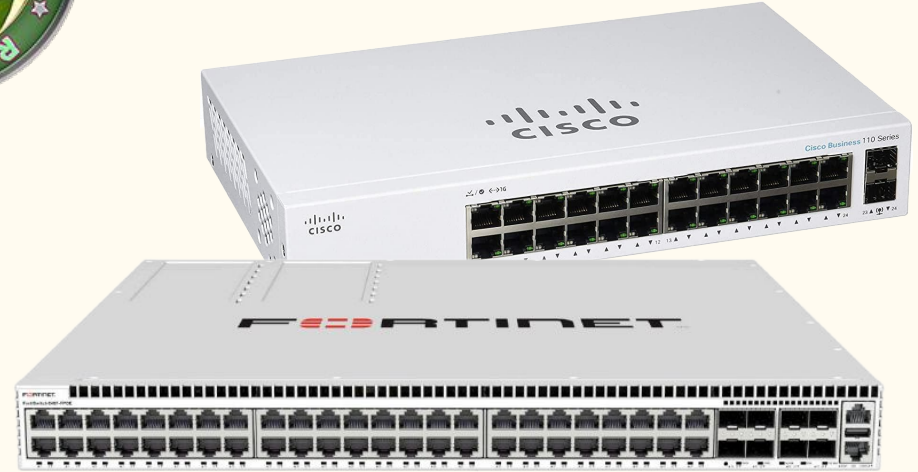
What do I need?

- Computers. duh



What do I need?

- Network. duhh...?



What do I need?

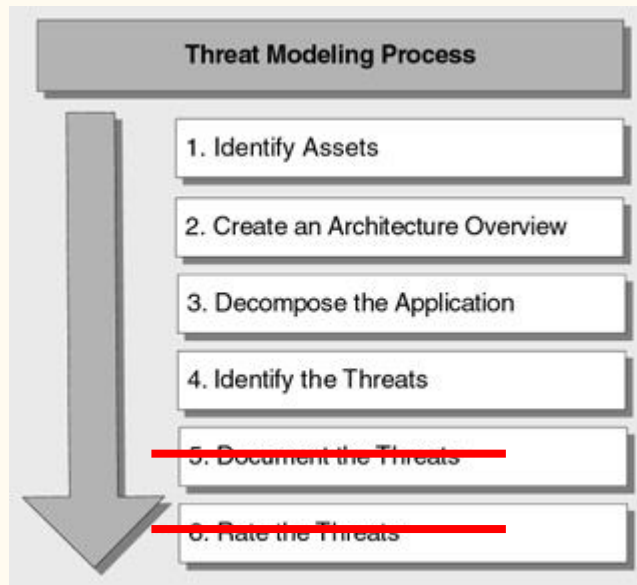
- Understanding how to do Resource Planning!
- For example:

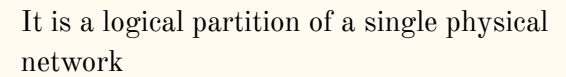


VM name	vCPUs	RAM	Storage	Resources
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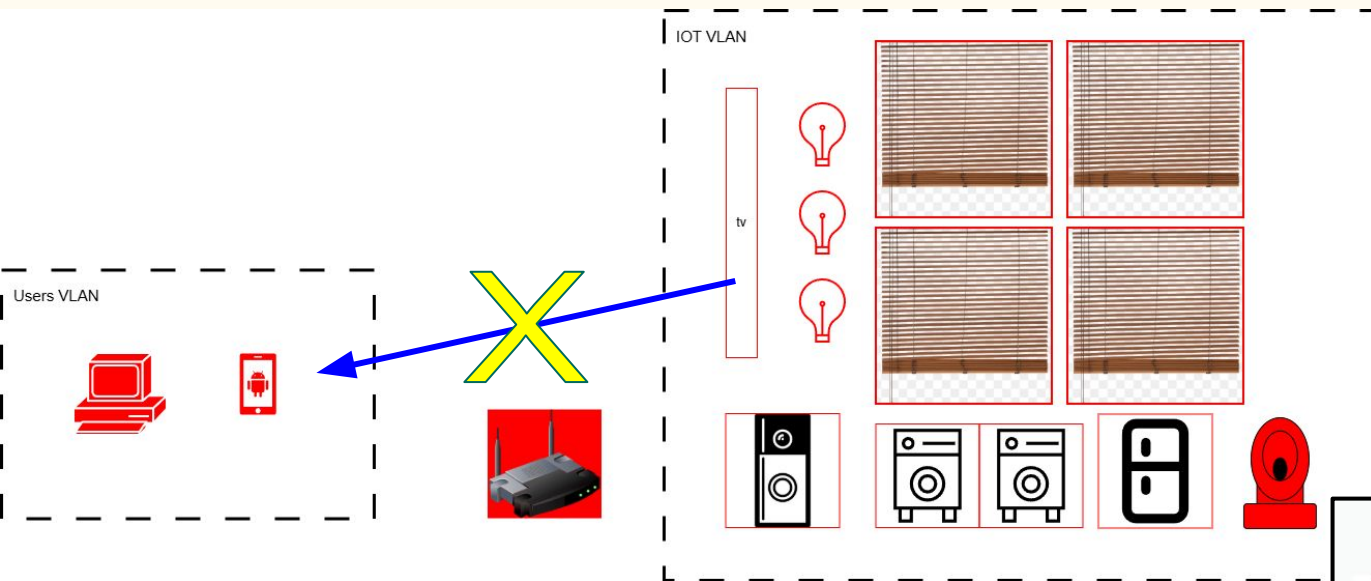
Threat Modeling

- A structured process used to identify, prioritize, and address potential security threats and vulnerabilities in a system before an adversary can exploit them.
- **1. Identify Assets:** Determine exactly what you are trying to protect (e.g., private data, media pools, smart home controls).
- **2. Create an Architecture Overview:** Map out the physical hardware, software stacks
- **3. Decompose the Application:** Break down the environment to understand data flows and trust boundaries
- **4. Identify the Threats:** Assume an asset is compromised. What impact am I looking? How do I reduce that impact?



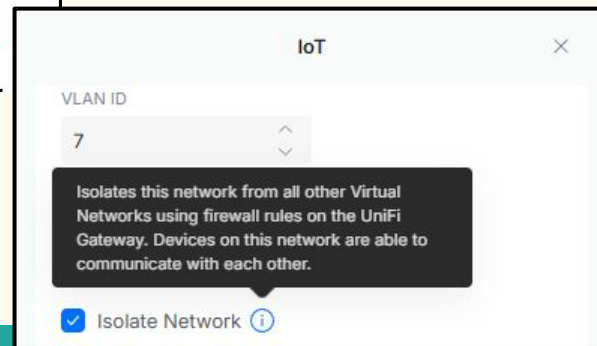


Average household today

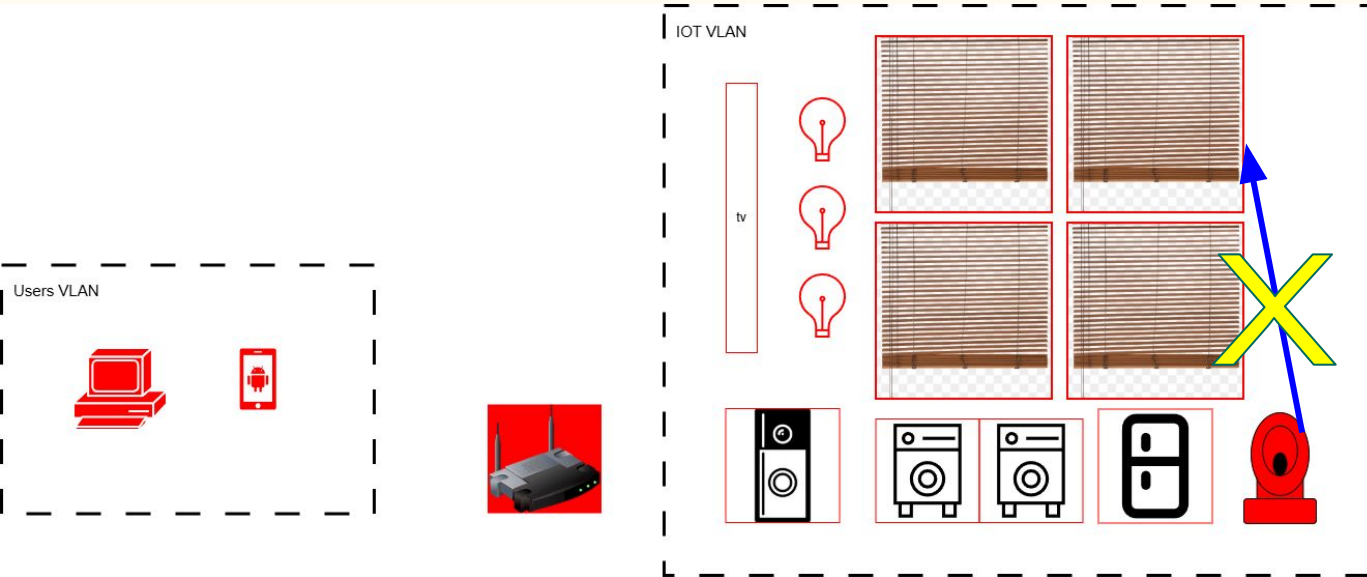


IOT devices should not:

1. actively probe other devices in in other VLANs



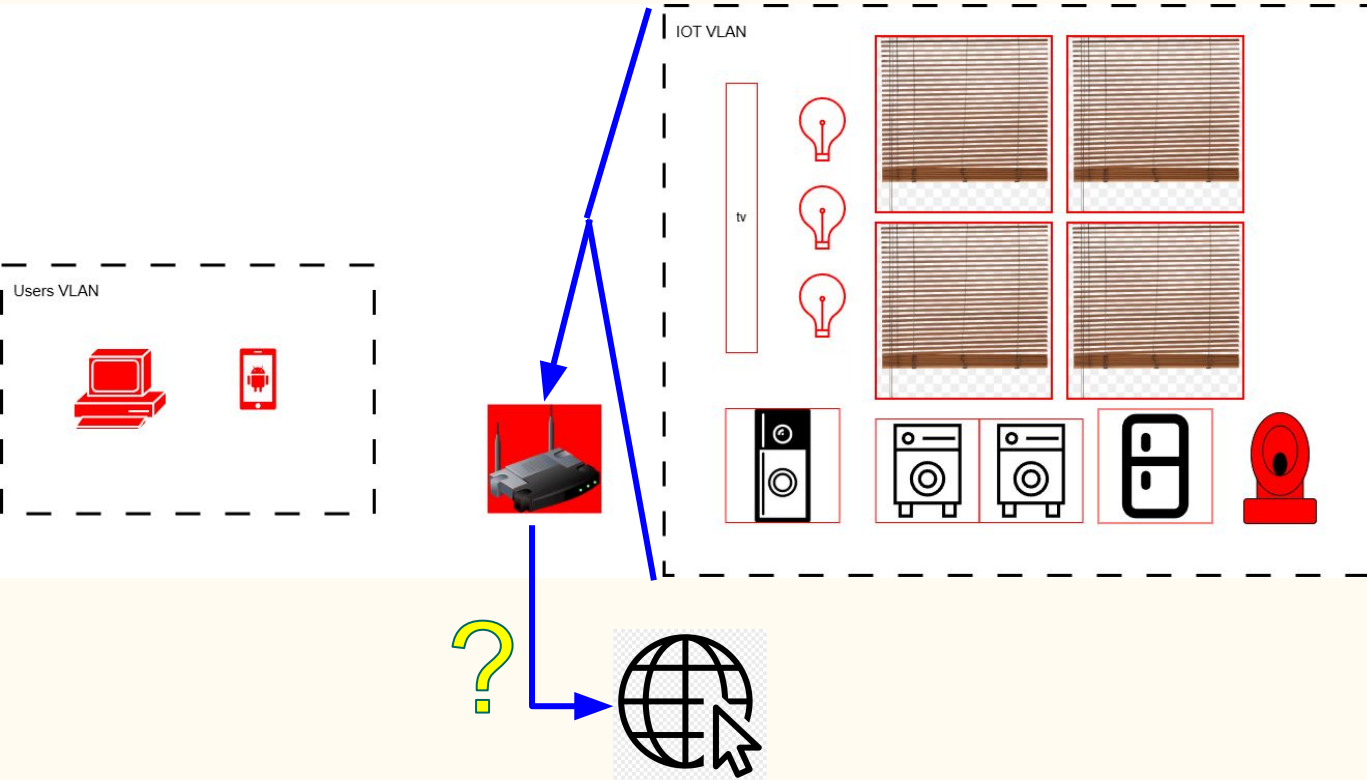
Average household today



IOT devices should not:

2. actively probe other devices within IOT VLANs

Average household today



Should IOT devices
connect to the
Internet?

Tuya still uses Wi-Fi
heavily

Depends on...

- Zigbee/Z-Wave
/Threads
- Matter/ Hubs

IoT Transport Protocols

Quick segue

Not the focus of this workshop, but
its a disservice to not even mention
it...

Why not Wi-fi-based IOT?

1. Security impacts

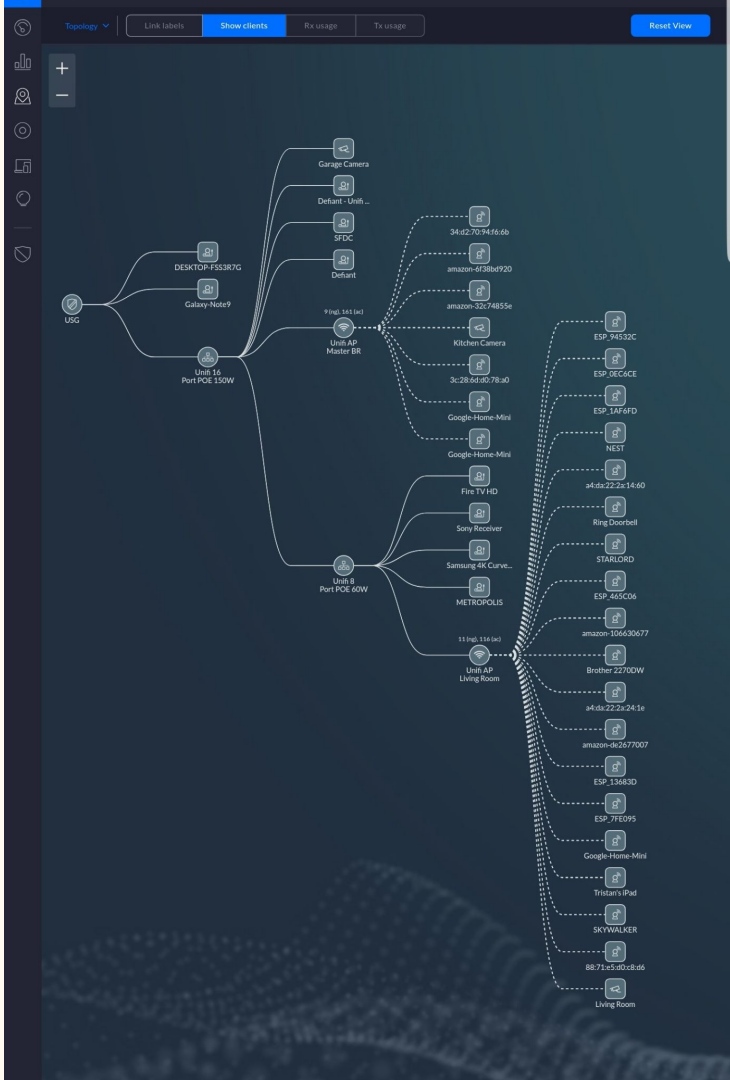
- a. Zombie botnets → attack targets unbeknownst to you
 - i. IOT devices connected to Internet
 - ii. Hardcoded credentials for telnet/ssh
 - iii. Mirai Botnet campaigns (2016)
 - iv. Botnet attacks report
- b. Domain takeovers (see report)
 - i. Malicious actors get sensitive data
 - ii. Can be used to push exploits to endpoint devices (supply chain)
- c. Linux OS with direct network probing capability

2. Network congestion

- Occupies registration in subnet
- Consumes traffic bandwidth

3. Maintenance headache

- Have to change wifi creds for each IoT device individually
- Hubs could just ‘adopt’



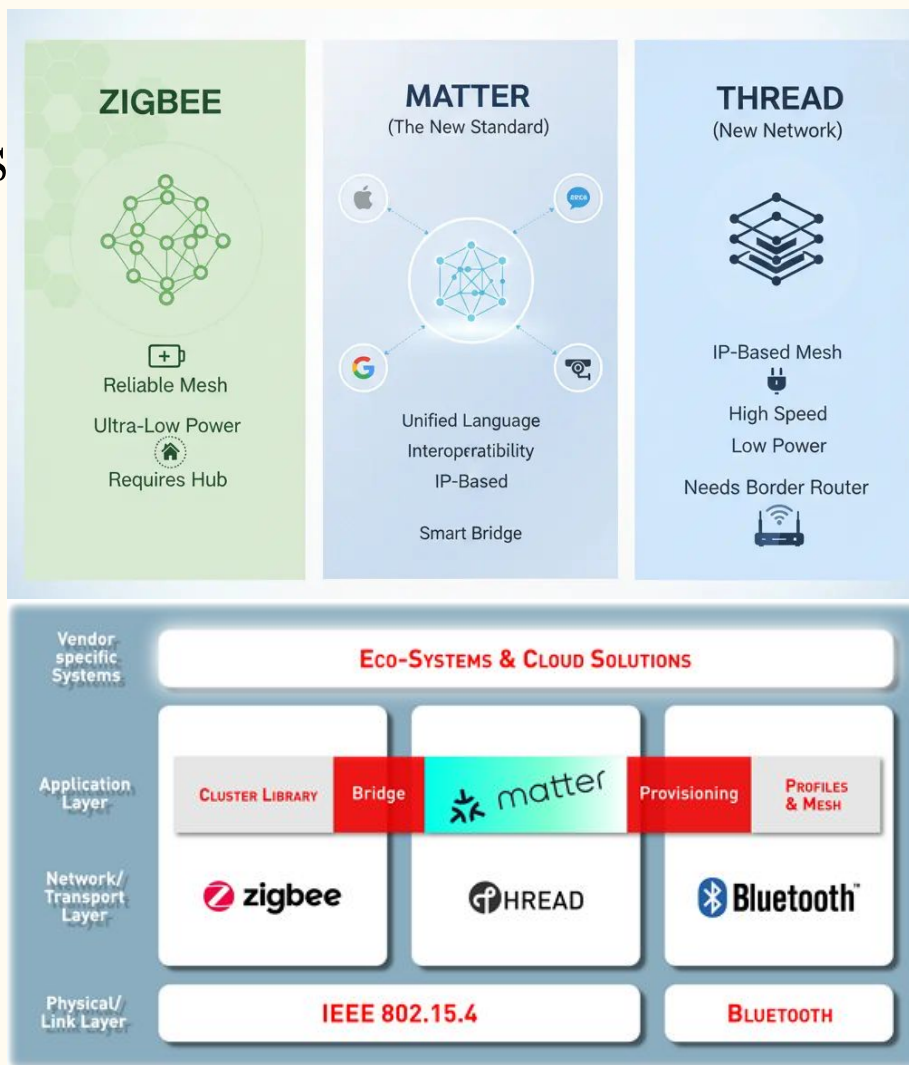
Are non-Wifi always good?

Lateral Movement through exploits in Philips Hue smart light bulbs (2020)

- a. Exploited Zigbee protocols
- b. But needed to gain privileges in Zigbee hub to do more
- c. <https://www.checkpoint.com/press-releases/the-dark-side-of-smart-lighting-check-point-research-shows-how-business-and-home-networks-can-be-hacked-from-a-lightbulb/>

IOT Protocols and Standards

- Zigbee and Threads
 - Network protocol (Layer 3)
 - Zigbee and Thread are both Mesh network (inter-device connectivity improves with more IOT devices)
- Matter
 - Application protocol (Layer 7)
 - Matter-over-Thread



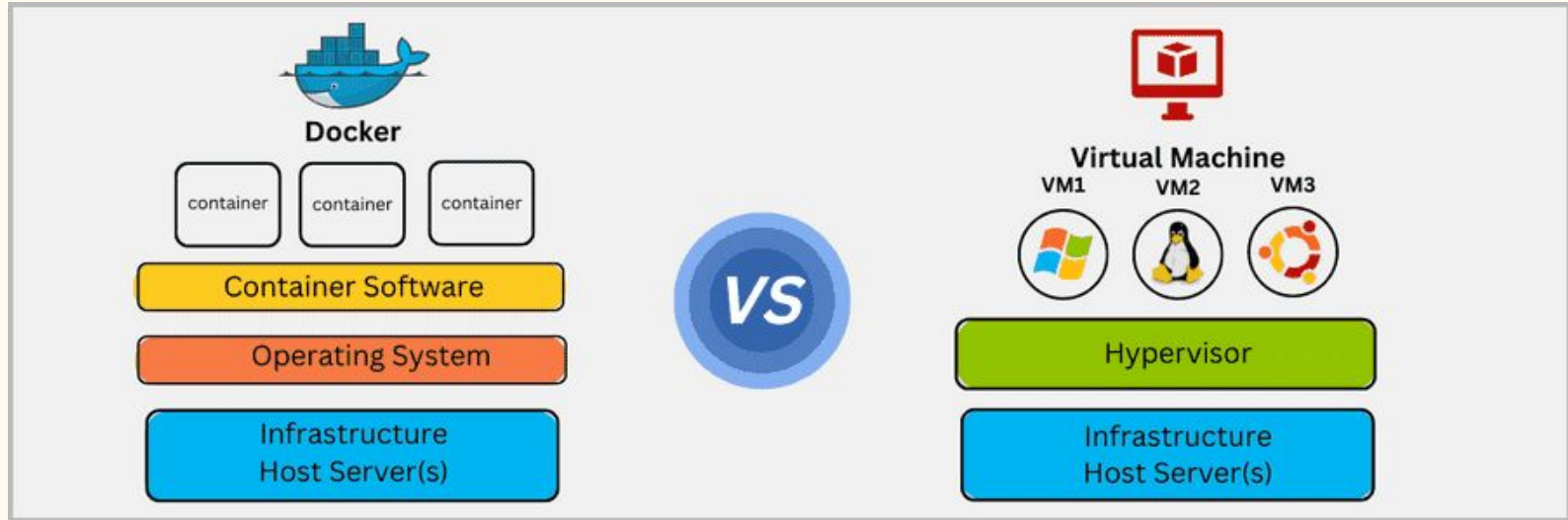
Part 2: Setting up Proxmox and DNS

Proxmox VE: The Homelab Engine

- **Type-1 Hypervisor**
- **Dual Virtualization:** Can run both VMs (KVM) and **Linux Containers (LXC)**
- **Enterprise for Free:** Zero paywalls. Advanced clustering and backup tools are 100% free.
- **IaC capable** using Hashicorp's tools

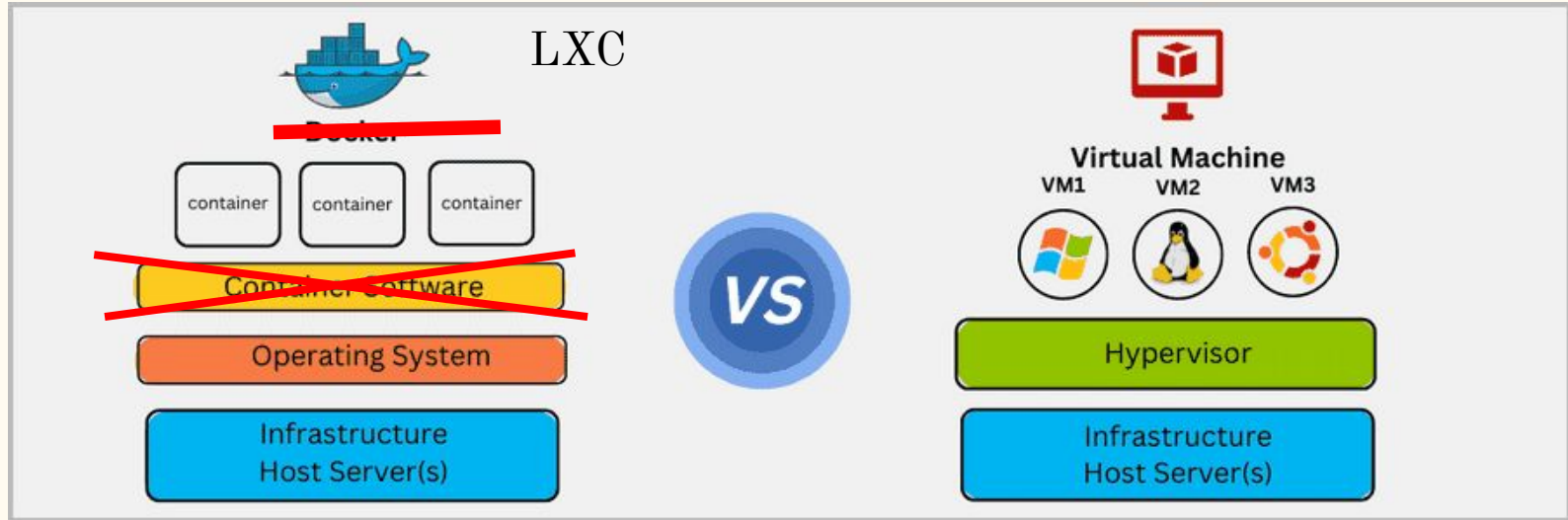


Remember this?



Then what is LXC?

LXC Container vs Docker container



LXC (Linux Containers) is like a lightweight OS without hypervisor, using kernel tricks, performed by the OS (proxmox)

- Like Docker, it shares the Host's kernel
- Unlike Docker, each container is a full-functioning OS (hence it has its own IP address)
- Unlike VM, each LXC is not virtualized (bare-metal)

Why Build your own DNS Server

- Low barrier to entry - Minimal hardware requirements
 - Raspberry Pi, old laptops, tiny VM
 - E.g. Pi-hole only requires 512 MB of RAM, 2GB of disk space
- Immediate, tangible benefits
 - **Network-wide** ad blocking
 - **Custom domain names** for homelab (e.g. truenas.home instead of 192.168.0.50)
- DNS query logging
- Increased security and privacy
 - DNS over HTTPS (DoH)
 - DNS over TLS (DoT)
 - DNS over QUIC (DoQ)
 - QUIC = HTTP3



Lab Time!

45 mins

Refer to the lab guide

Lab2 Proxmox

Part 3: Log Monitoring



Why centralise logs?

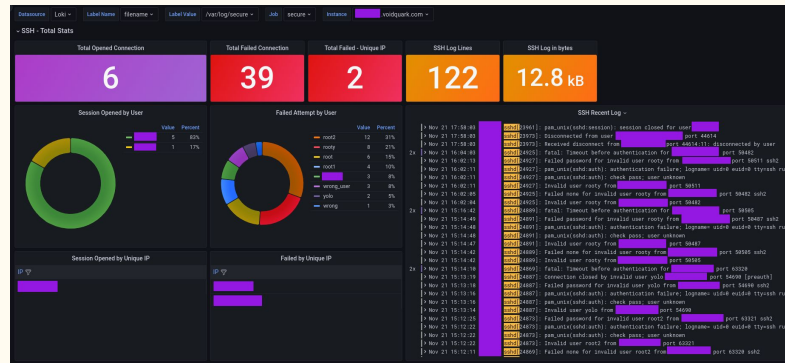


```

root@adc1c1:~# egrep "Failed\[failure\]" /var/log/auth.log
Dec 5 21:39:17 adc1 sshd[41458]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0
tty=ssh ruser= rhost=192.168.1.3 user=root
Dec 5 21:39:20 adc1 sshd[41458]: Failed password for root from 192.168.1.3 port 37362 ssh2
Dec 5 21:39:23 adc1 sshd[41458]: Failed password for root from 192.168.1.3 port 37362 ssh2
Dec 5 21:39:28 adc1 sshd[41458]: Failed password for root from 192.168.1.3 port 37362 ssh2
Dec 5 21:39:28 adc1 sshd[41458]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh
ruser= rhost=192.168.1.3 user=root
Dec 5 21:39:41 adc1 sshd[41469]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0
tty=ssh ruser= rhost=192.168.1.3 user=tecmint
Dec 5 21:39:44 adc1 sshd[41469]: Failed password for tecmint from 192.168.1.3 port 37364 ssh2
Dec 5 21:40:18 adc1 sshd[41491]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0
tty=ssh ruser= rhost=192.168.1.245 user=admin
Dec 5 21:40:21 adc1 sshd[41491]: Failed password for root from 192.168.1.245 port 52882 ssh2
Dec 5 21:40:24 adc1 sshd[41491]: Failed password for root from 192.168.1.245 port 52882 ssh2
Dec 5 21:40:30 adc1 sshd[41491]: Failed password for root from 192.168.1.245 port 52882 ssh2
Dec 5 21:40:30 adc1 sshd[41491]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh
ruser= rhost=192.168.1.245 user=root
Dec 5 21:40:42 adc1 sshd[41506]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0
tty=ssh ruser= rhost=192.168.1.245 user=admin
Dec 5 21:40:42 adc1 sshd[41506]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_AUTH_
ERROR, PAM error: PAM_AUTH_ERR (7), NTSTATUS: NT_STATUS_LOGON_FAILURE, Error message was: Logon fail
ure
Dec 5 21:40:45 adc1 sshd[41506]: Failed password for admin from 192.168.1.245 port 52884 ssh2
root@adc1c1:~# _

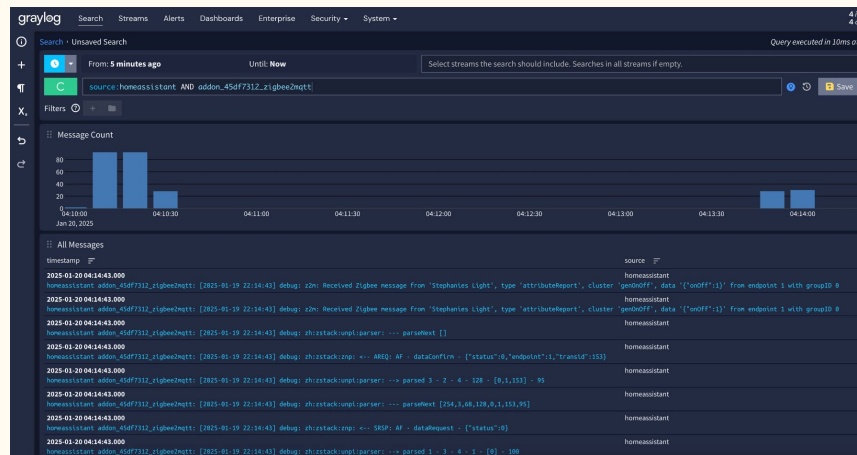
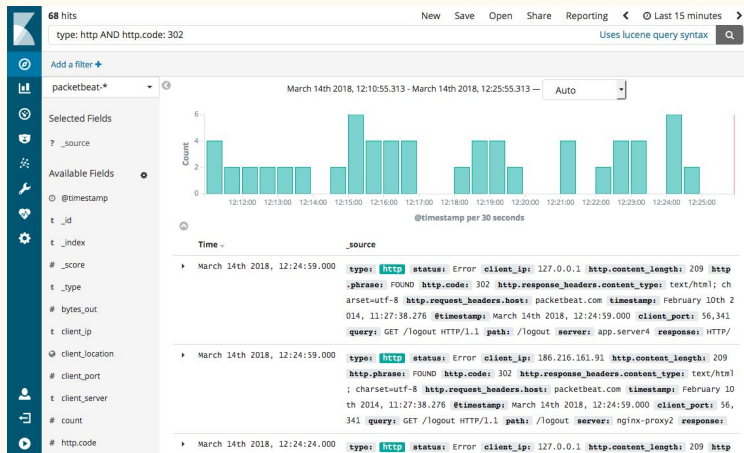
```

SSH into individual servers to grep auth.log to find login failures



Central dashboard that displays SSH login failures and stats from all hosts, with alert support

Logging Options for Homelabs



ELK (Elastic + Logstash + Kibana)

Graylog

High Resource Usage!

What is...

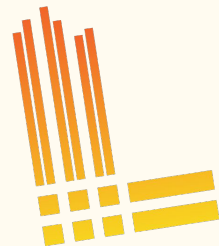
- **Grafana**

- **Dashboard layer** - Query, visualise and alert on metrics, logs and traces
- Connects many data sources - Prometheus (Metrics), Loki (Logs)



- **Loki**

- **Where logs land and are stored** - e.g. /var/log, docker logs etc.
- Query them using LogQL directly in Grafana

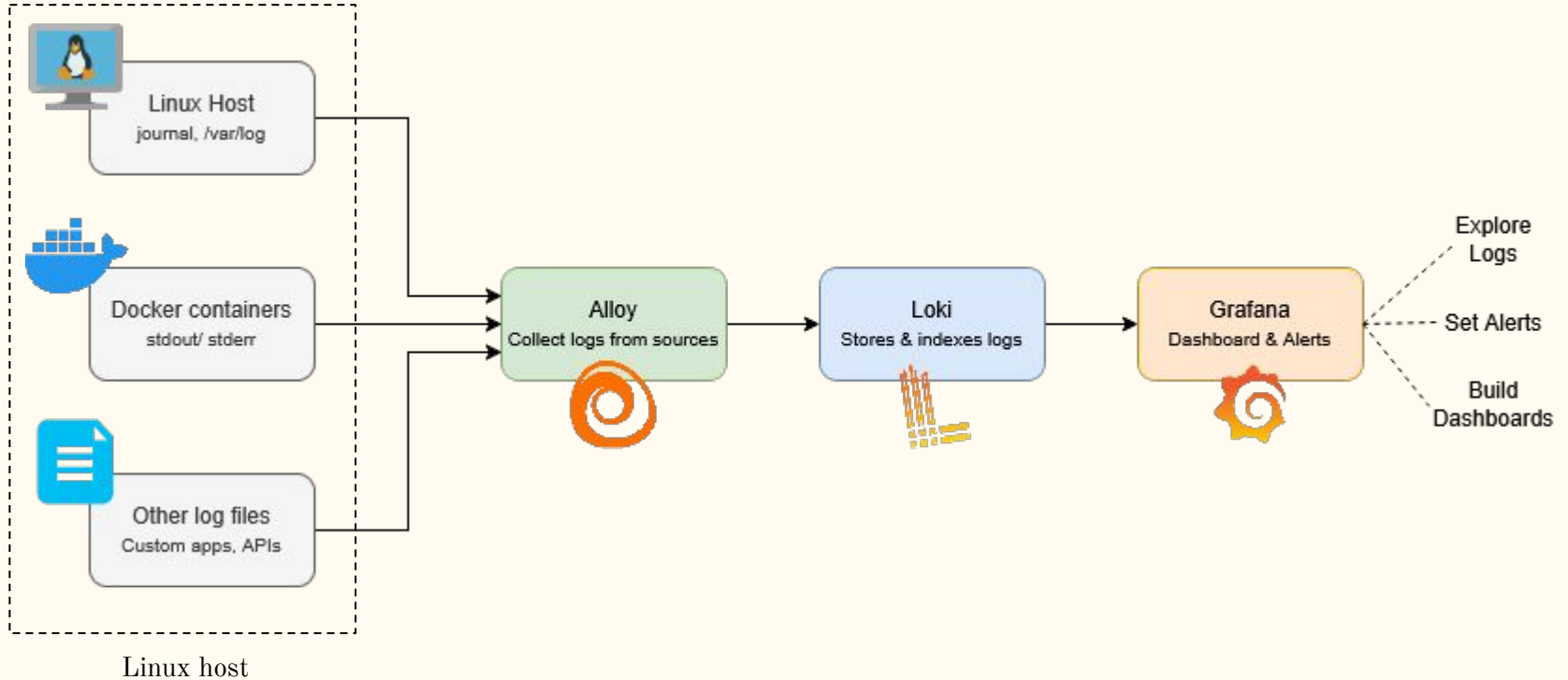


- **Alloy**

- **Collector Agent** - Agent installed on log sources to forward them to Loki
- Is a unified agent that also forwards metrics to Prometheus, etc.

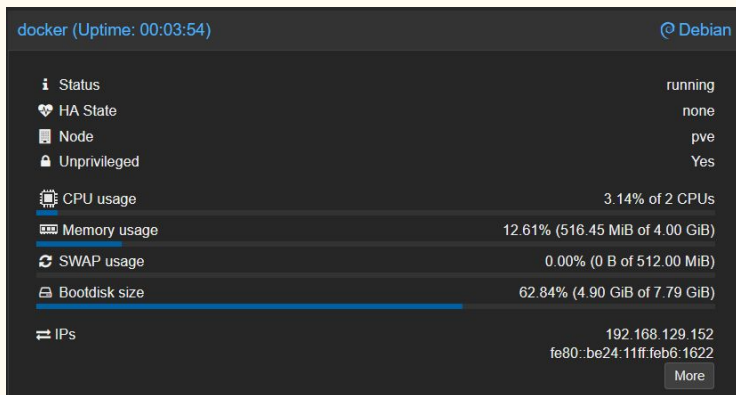


How they come together

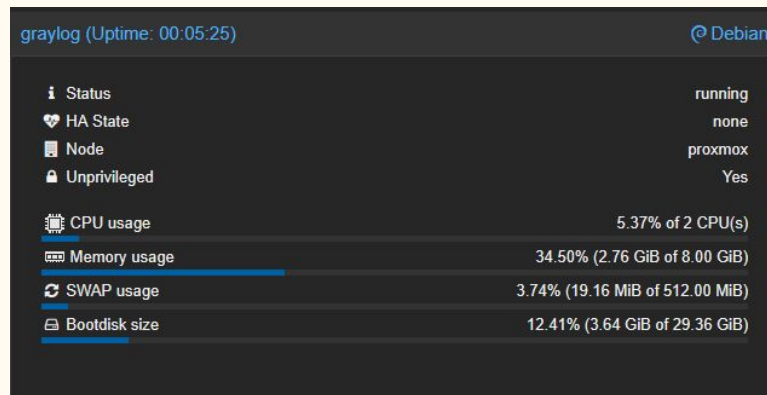


Why did I choose the Grafana stack?

At idle:



Our workshop stack only needs
516.45 MiB of RAM combined



Graylog LXC uses 2.76
GiB of RAM!

Lab Time!

45 mins

Refer to the lab guide

Lab3_Grafana

Part 4: Whats next

—

Exposing homelab to Internet

Suppose you have installed other webapps that you want to access anywhere outside of your local domain...

- Own “Photos” app
- Malware sandbox, etc.

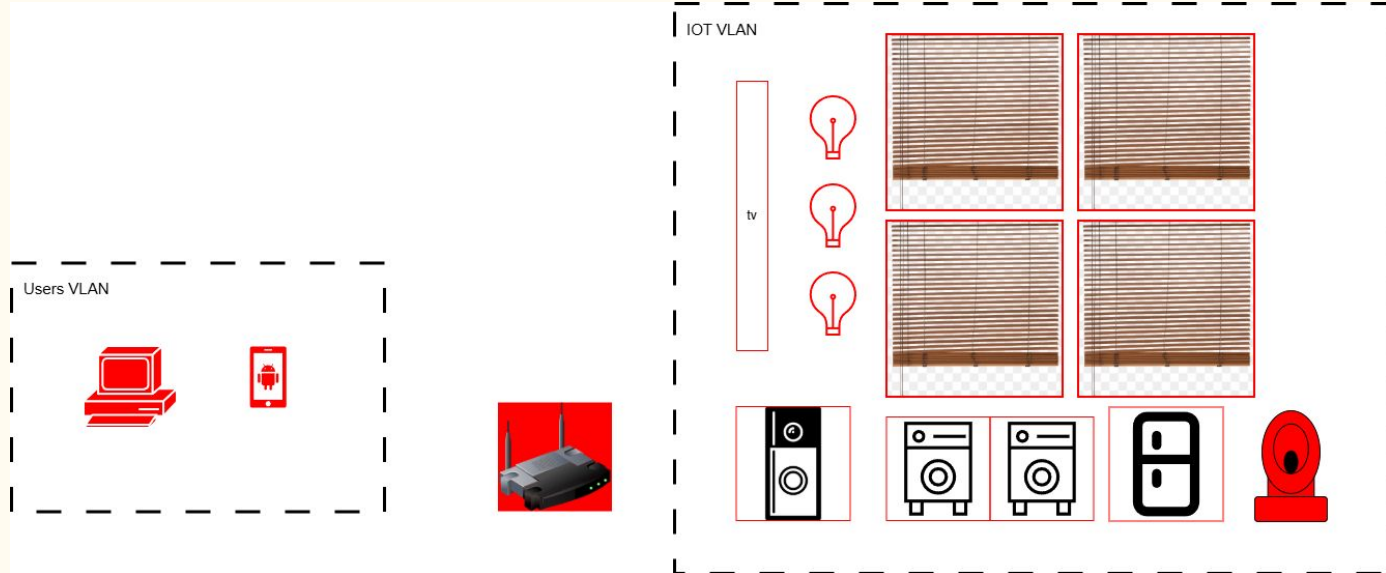
What should we do?



Exercise... Exposing your homelab to Internet

Recall our simplified threat modeling exercise we did earlier

This is before we set up our homelab.



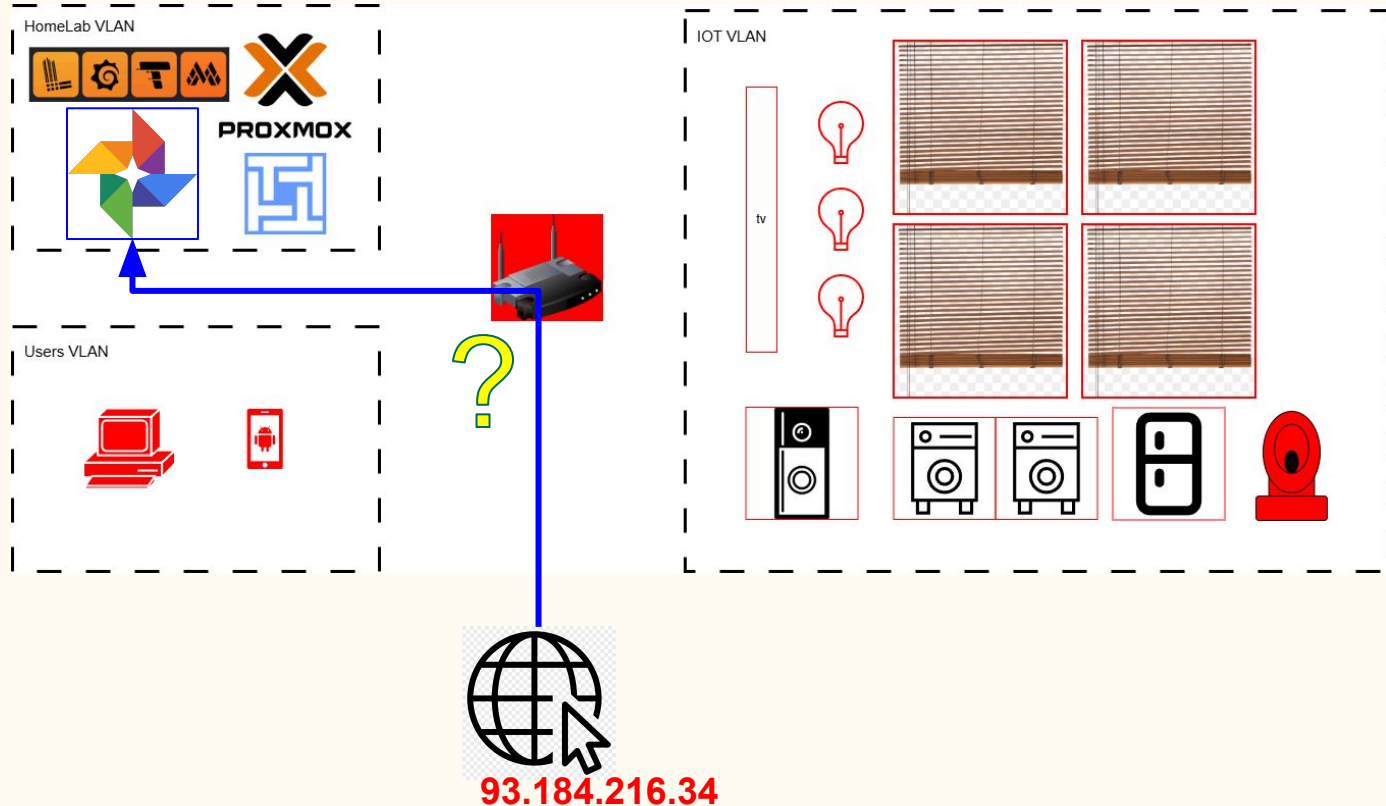
Example of our public IP 93.184.216.34

Exercise... Exposing your homelab to Internet

Now, when we set it up at home, we also need a Homelab VLAN

You might also want to create a Management VLAN so that only some users can access Proxmox UI and Technitium DNS WebUI

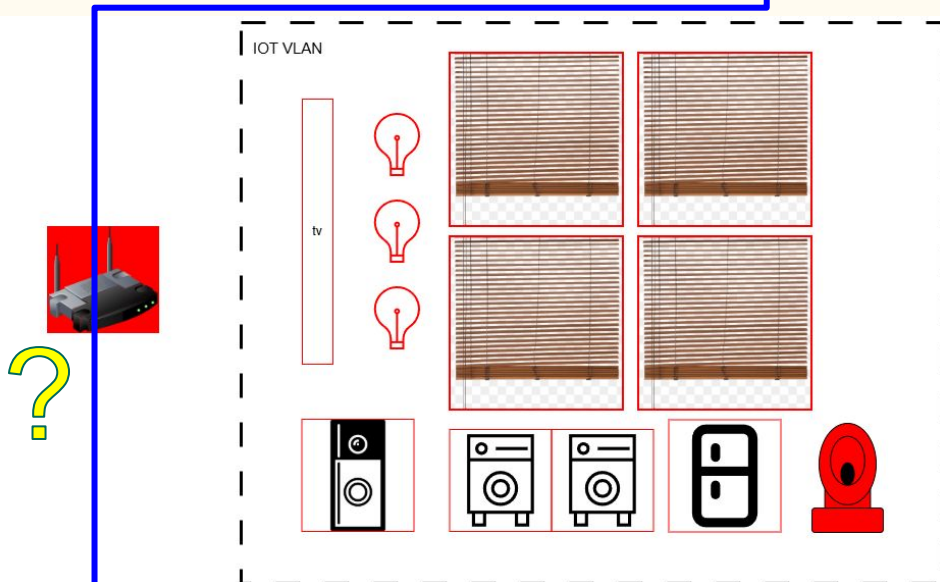
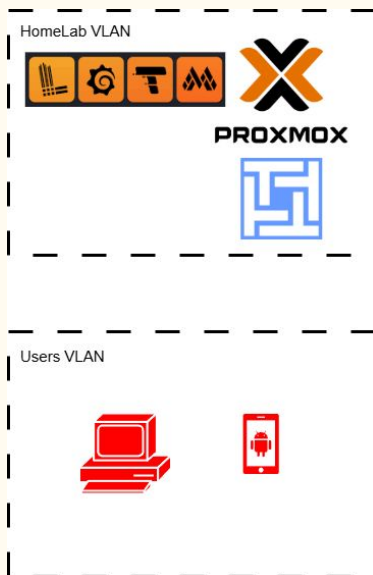
How should we expose our services to the Internet?



Exposing homelab to Internet. DMZ?

What is Demilitarized Zone (DMZ)?

- Standard VLAN with default policy to prevent devices in DMZ to probe internal network
- Allow external → gateway → DMZ
- require **port forwarding**



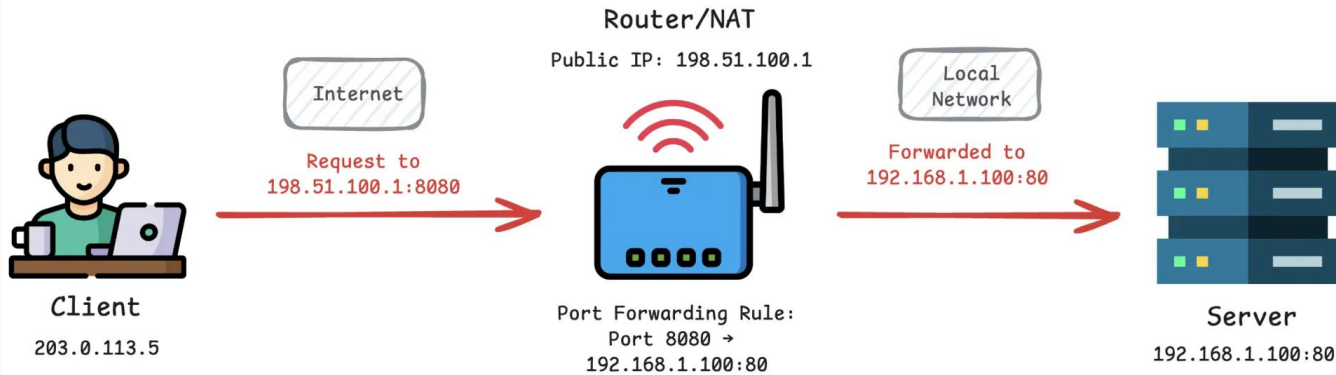
Allow DHCP	Firewall	Allow	UDP	DMZ	Any	Gateway	Any
+							
Allow DNS	Firewall	Allow	TCP/UDP	DMZ	Any	Gateway	Any
Allow ICMP	Firewall	Allow	ICMP	DMZ	Any	Gateway	Any
Allow ICMPv6	Firewall	Allow	ICMPv6	DMZ	Any	Gateway	Any



93.184.216.34

What is port forwarding?

Port Forwarding Concept



1. Client sends request to public IP (198.51.100.1) on port 8080
2. Router receives the request and checks port forwarding rules
3. Router forwards the request to internal server (192.168.1.100) on port 80
4. Server processes the request and sends response back through the same path

https://pinggy.io/blog/how_to_set_up_port_forwarding_even_behind_cgnat/

Create Policy

☐ Firewall ☐ Route ☐ QoS ☐ NAT
☐ DNS ☐ ACL ☒ Port Forwarding

Name

WAN Interface
☒ WAN1 ☐ WAN2 ☐ All WANs

WAN IP Address ⓘ
DHCP

WAN Port ⓘ
e.g. 1-10,11,12

Update Forward Port

From ⓘ
☒ Any ☐ Limited

Forward IP Address ⓘ
IPv4 Address

Select Device

Forward Port ⓘ
e.g. 1-10,11,12

Protocol
☒ TCP/UDP ☐ TCP ☐ UDP

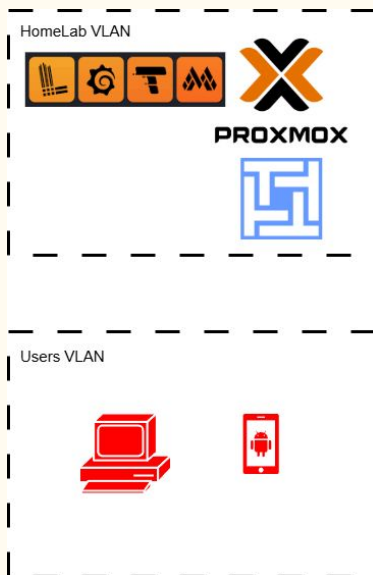
☐ Syslog Logging ⓘ

Add

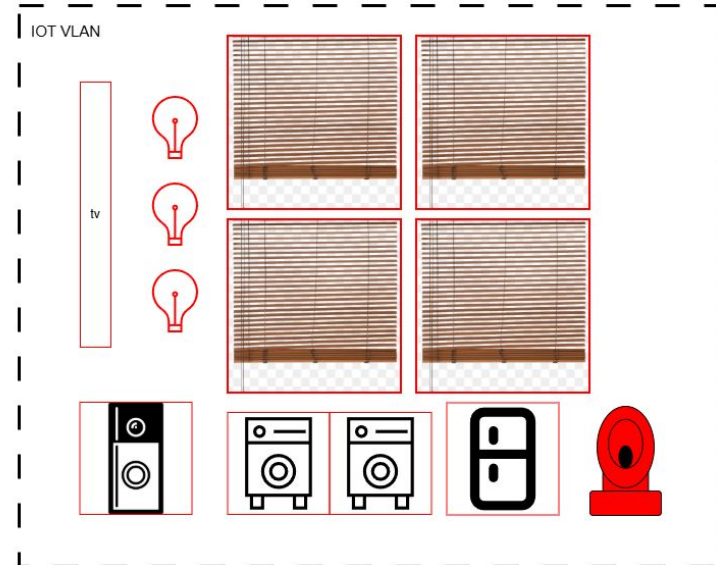
Exposing homelab to Internet. DMZ?

Why not DMZ?

1. Exposes public IP address (danger)
2. Public IP address resets periodically (**inconvenience**)
3. (with no port forwarding DMZ) exposes all services. Increase Attack surface



DMZ VLAN



Allow DHCP	Firewall	Allow	UDP	DMZ	Any	Gateway	Any
+							
Allow DNS	Firewall	Allow	TCP/UDP	DMZ	Any	Gateway	Any
Allow ICMP	Firewall	Allow	ICMP	DMZ	Any	Gateway	Any
Allow ICMPv6	Firewall	Allow	ICMPv6	DMZ	Any	Gateway	Any



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can we do better?

Exposing homelab to Internet. DMZ?

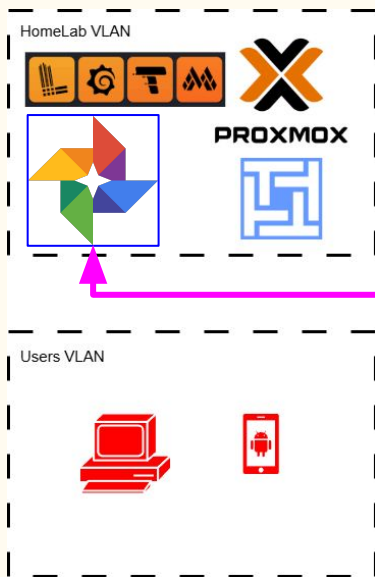
Instead of direct service, use a Proxy Manager.

Popular reverse proxies include:

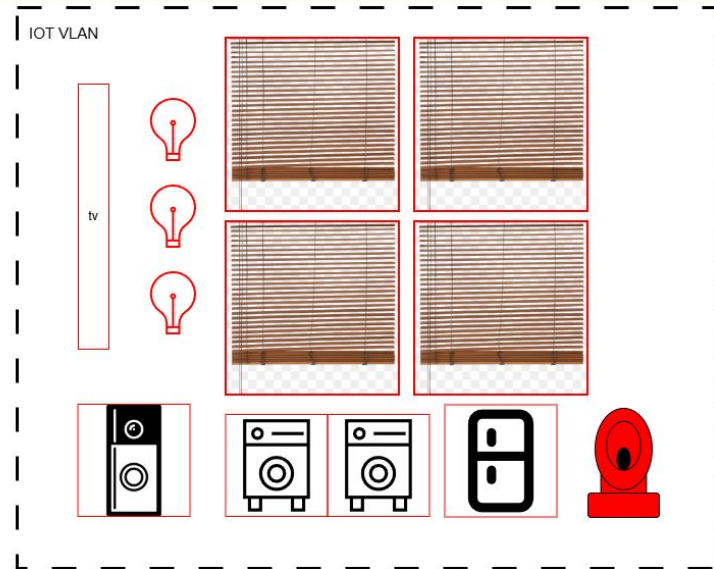
1. Nginx Proxy Manager (NPM)
2. Traefik
3. Caddy

These also often include **ACME** services

Automated Certificate Management Environment (ACME) for TLS certs



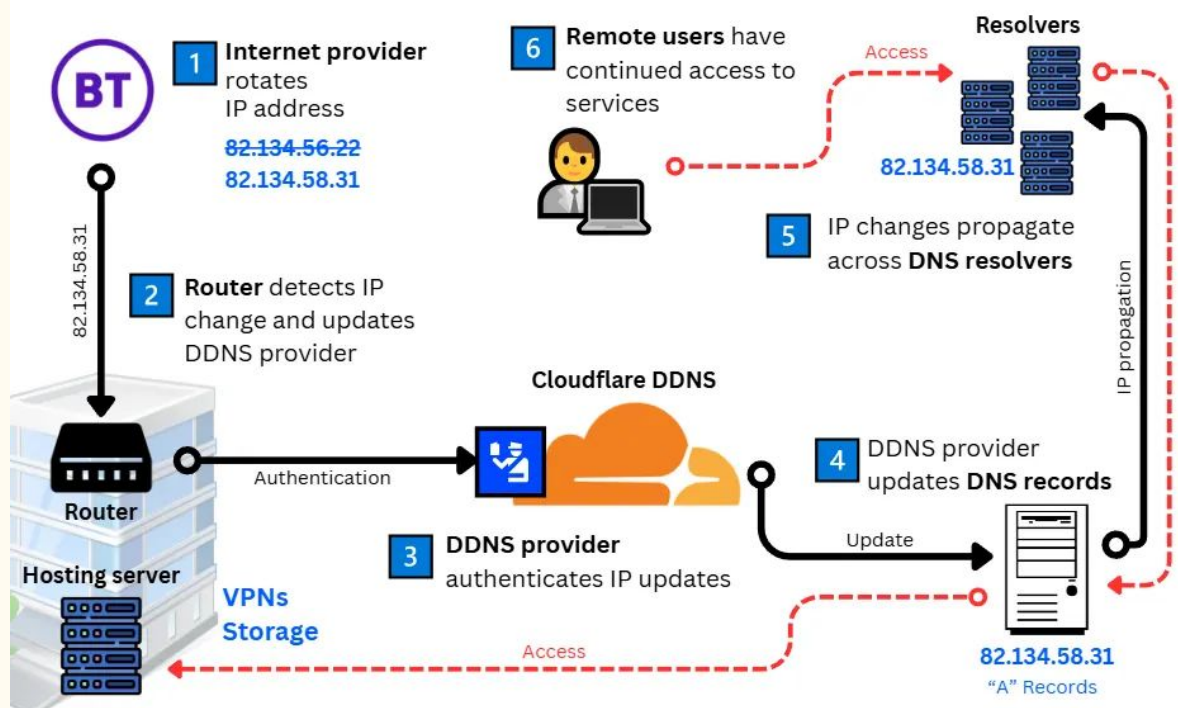
DMZ VLAN



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are we good yet?

Dynamic DNS (DDNS)



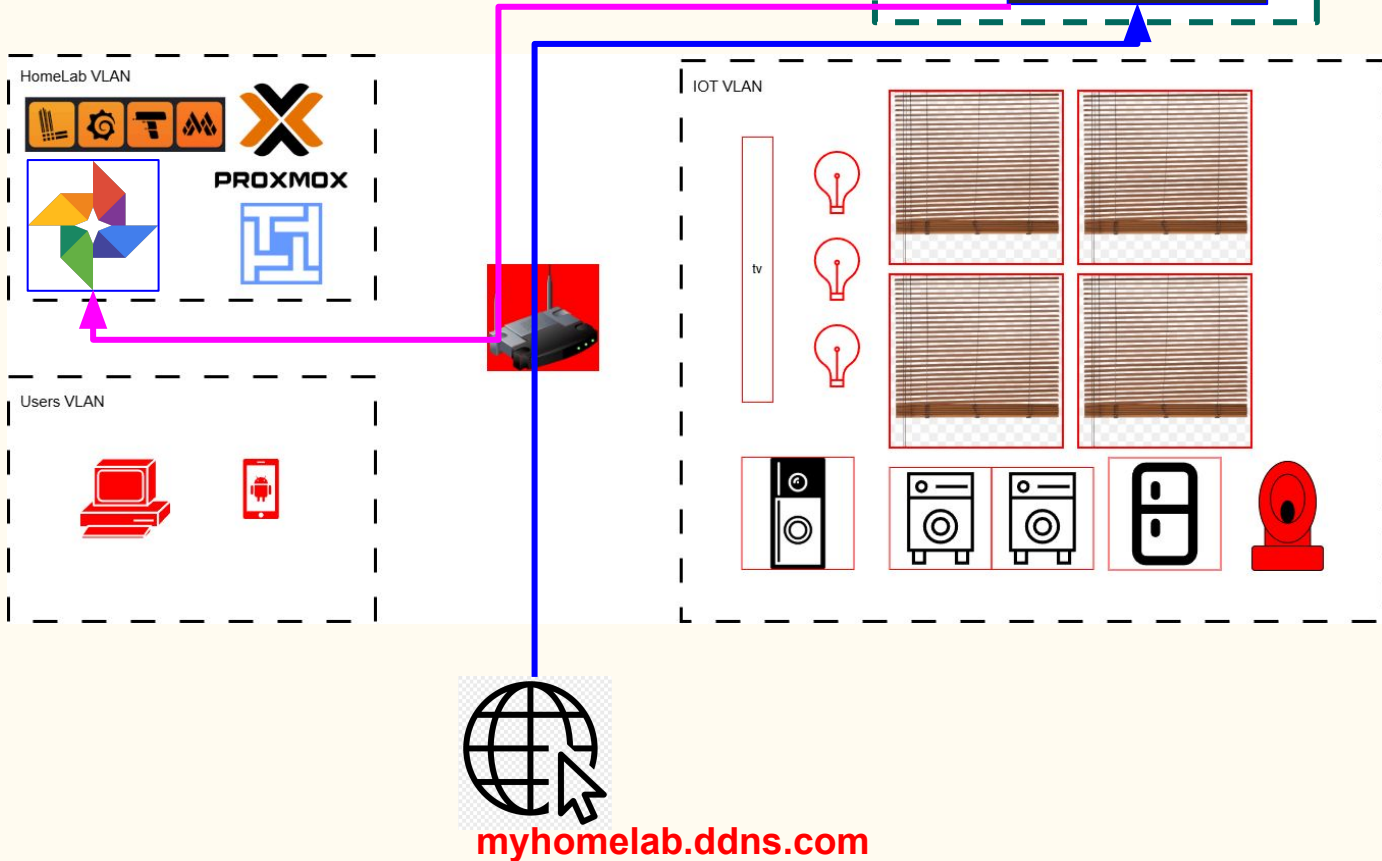
Even with a Reverse Proxy, our public IP address is still exposed.

Exposing homelab to Internet. DDNS?

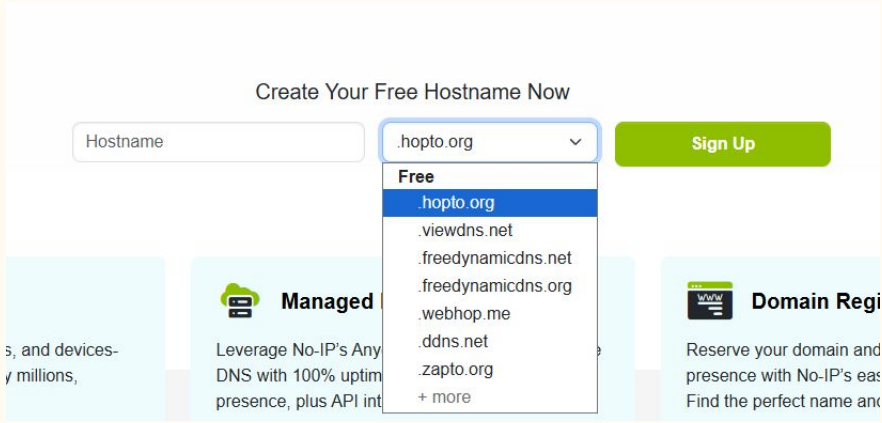
Setting up a Dynamic DNS lets our Router do the heavily lifting

Usually piggybacking on subdomains to resolve to an IP that is managed by your Router.

NOTE: some DDNS piggybacked domains do not have a good reputation because of their 'free'-ness and abused by threat actors, which could limit your ability to reach your homelab through DDNS if you are behind a firewall/NIPS.

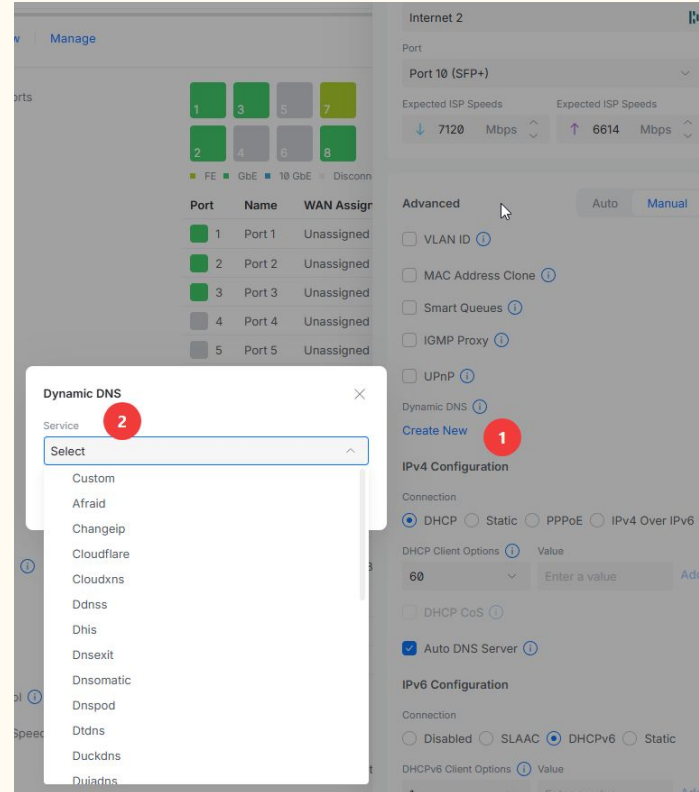


Dynamic DNS (DDNS)



1. Create your DDNS hostname

2. Set up DDNS in your router (easiest)



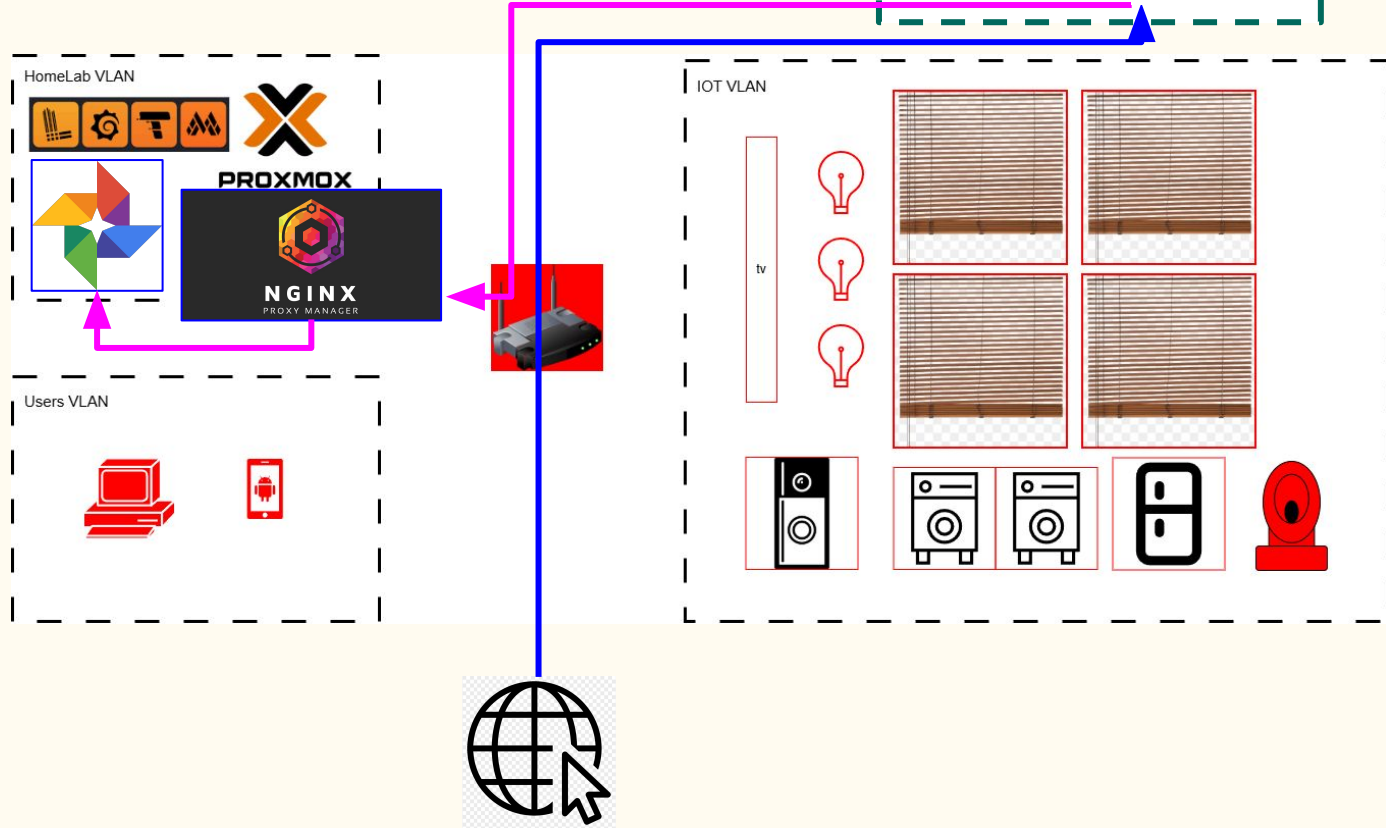
Exposing homelab to Internet. VPN (1)

Routers may include
VPN services, like Unifi
Teleport over WireGuard

Create policies to restrict
services allowed from the
VPN VLAN.

NPM/Traefik/Caddy
move to Homelab VLAN

Reverse Proxies are
always a good idea
(whitelist mentality)

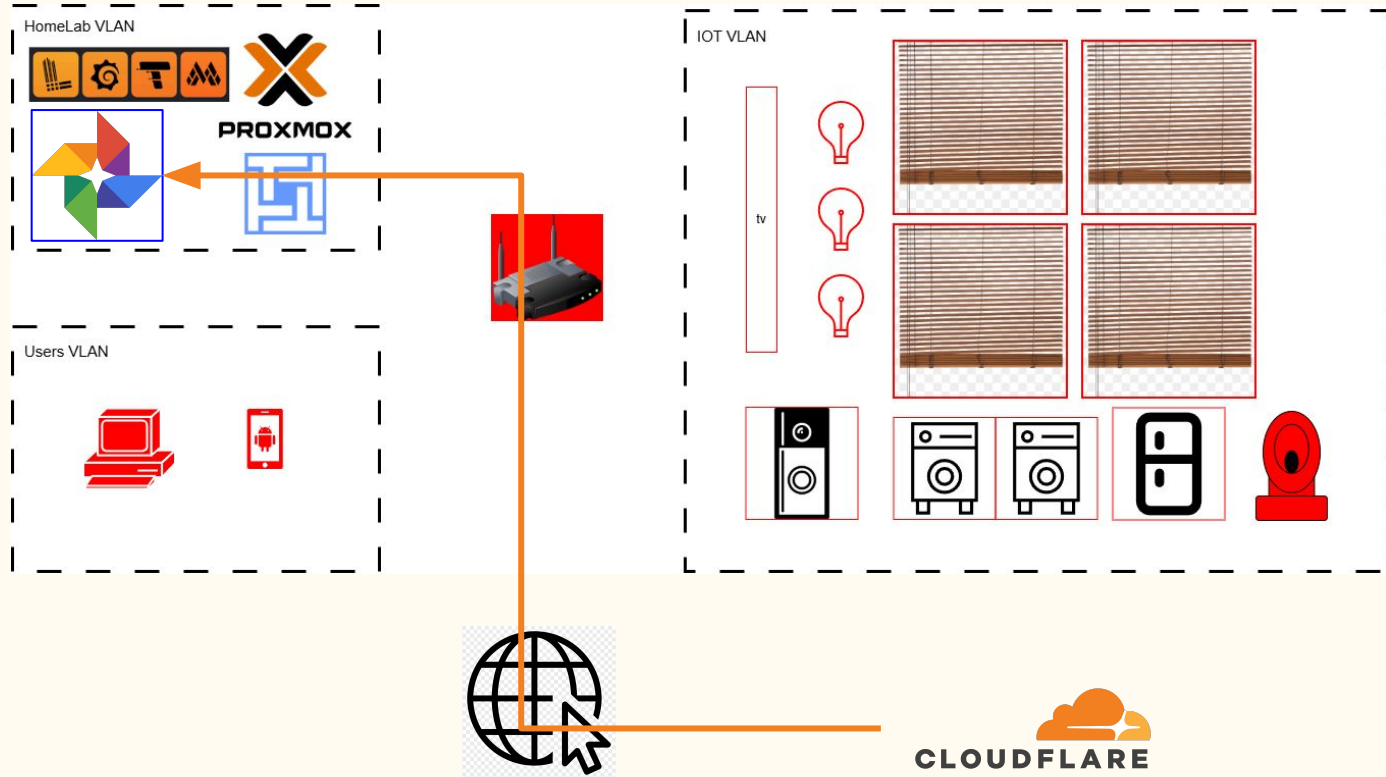


Exposing homelab to Internet. VPN (2)

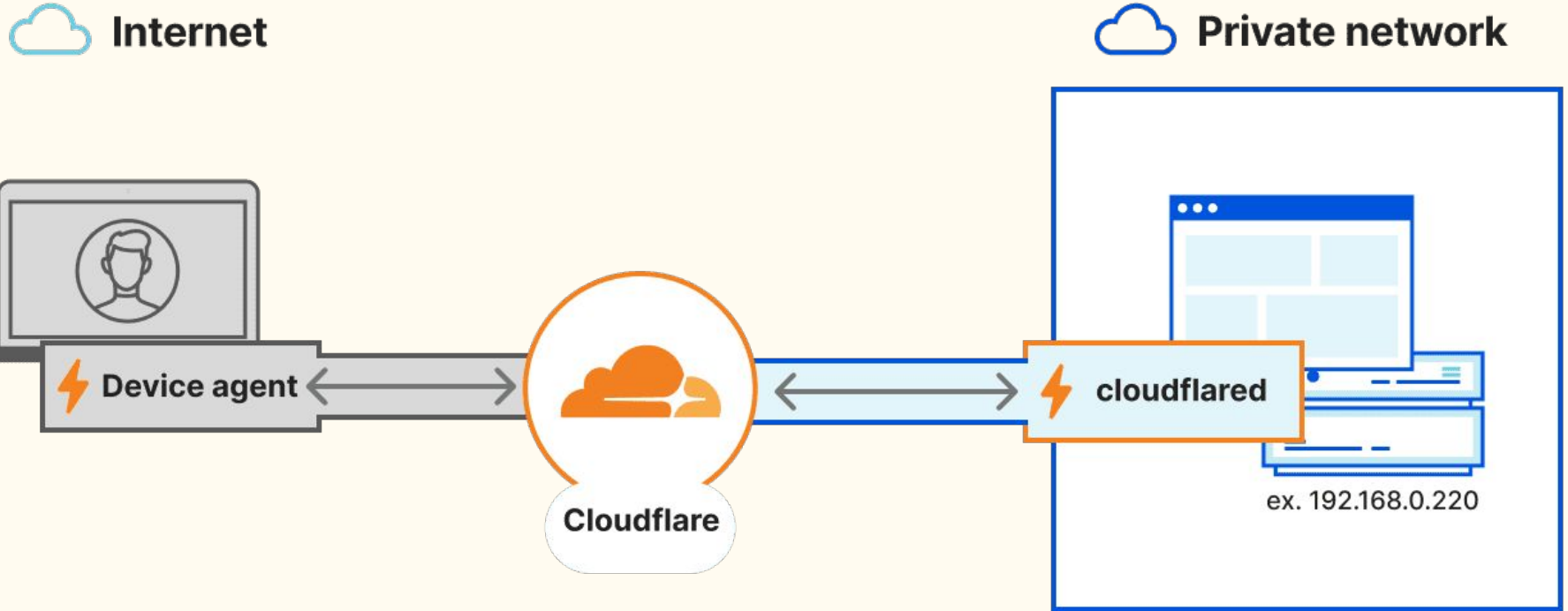
Cloudflare WARP is free, but no tunnel support

Cloudflare Zero Trust is a ~~paid~~ free service that allow private tunnel.

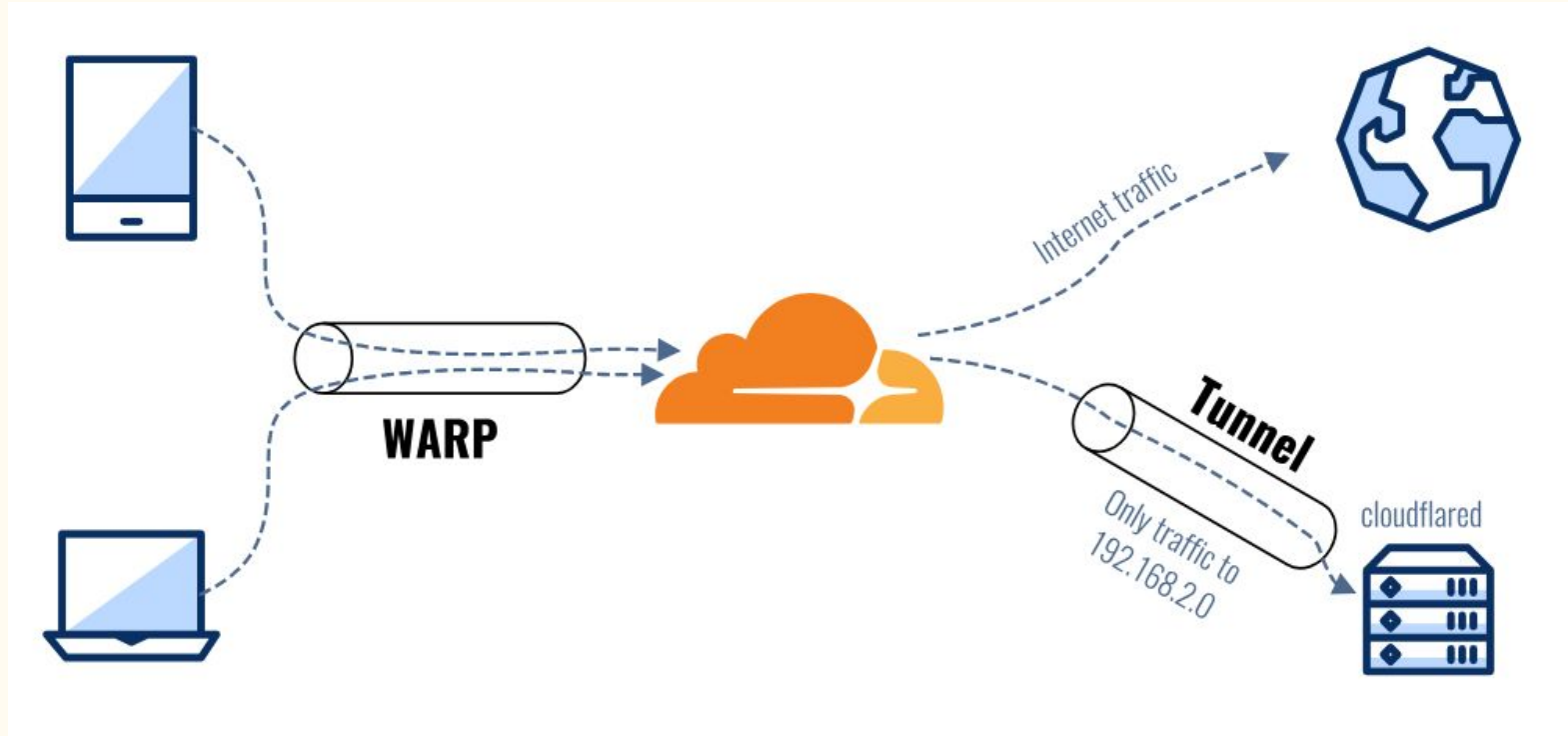
This is not a sponsored ad. If you want to read more about it, go [here](#)



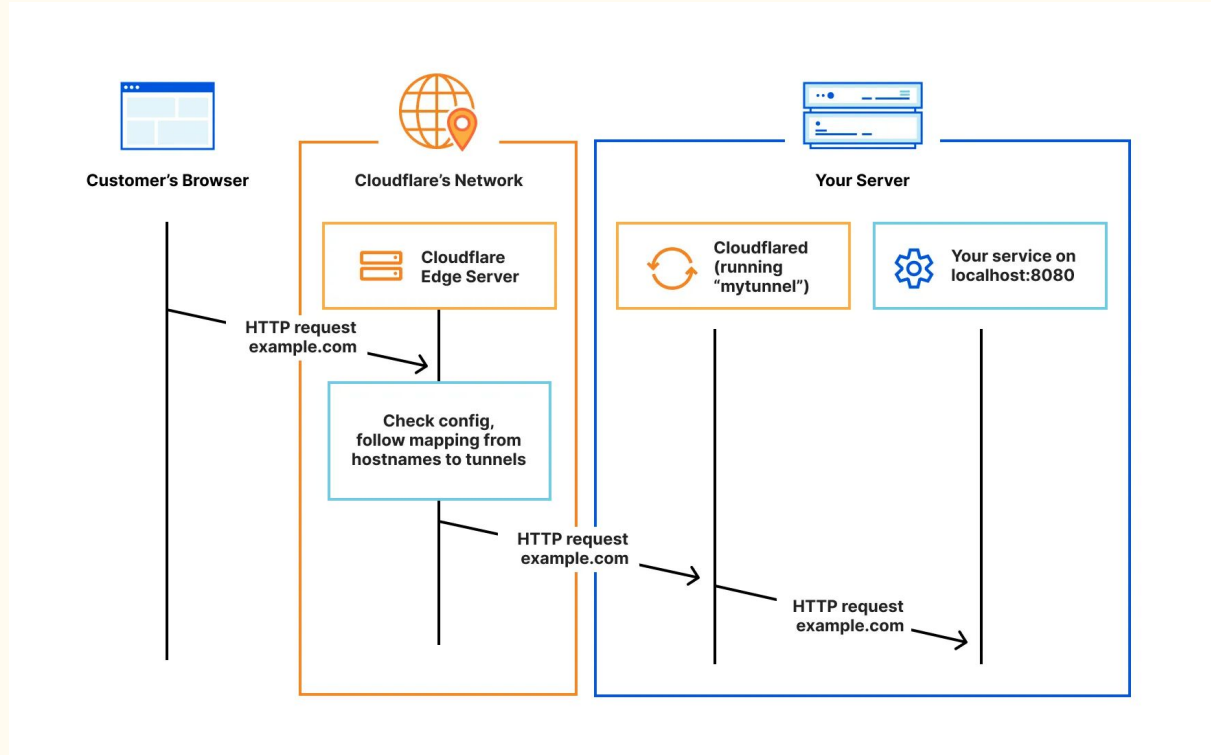
Cloudflare Zero Trust



Cloudflare tunneling



Cloudflare tunneling

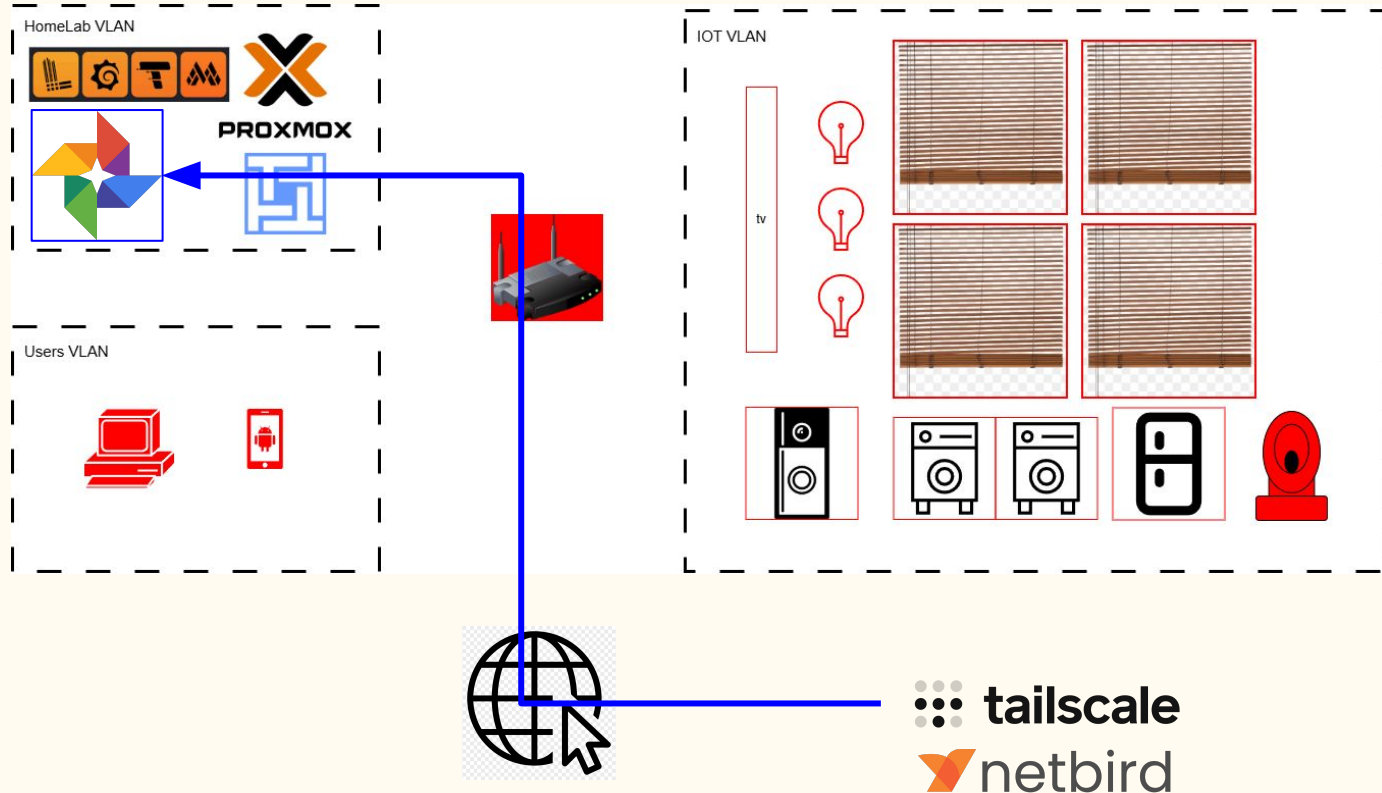


Exposing homelab to Internet. VPN (3)

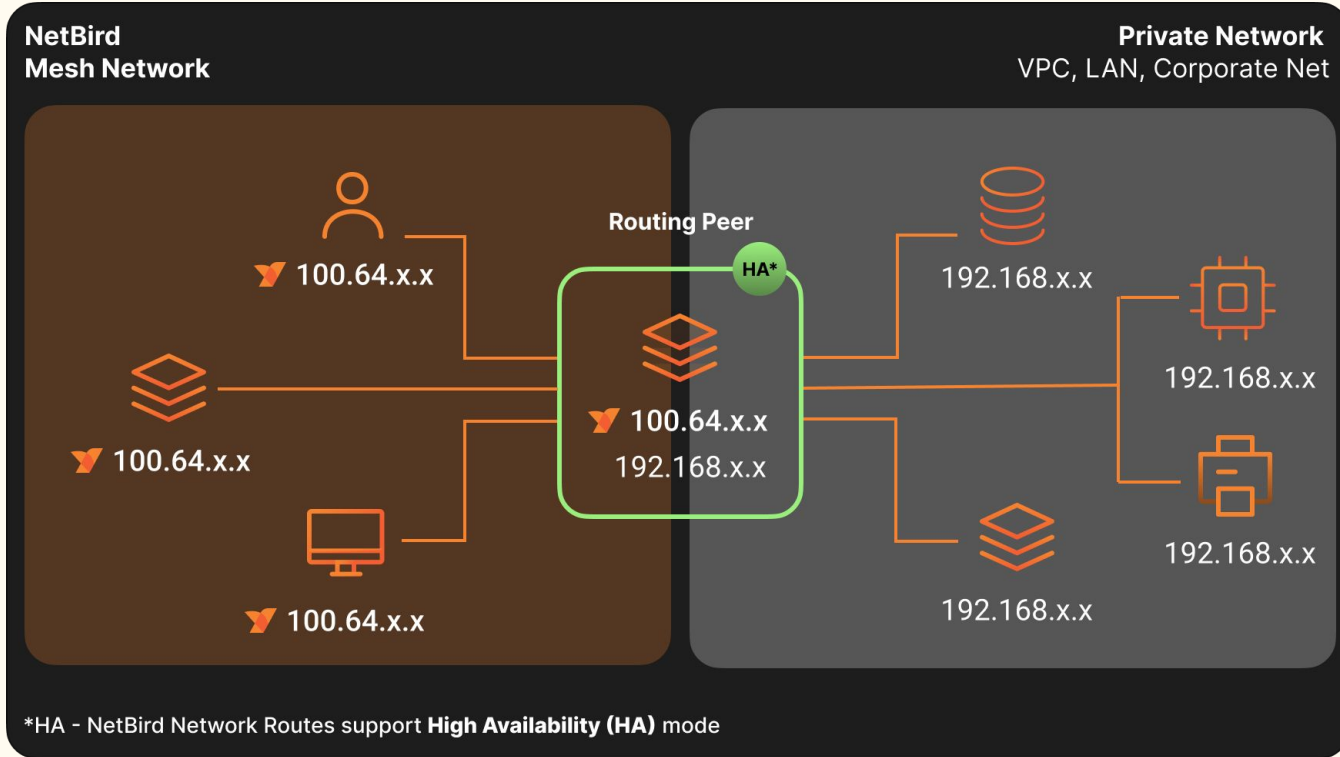
Tailscale and Netbird are the most important VPN services you will hear in the homelab community

They are both freemium products. Tailscale is older than netbird (thus may be considered more mature)

Netbird is a dark horse (IMO)

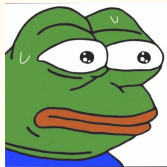


Tailscale/Netbird



Disclaimer over Exposing Your Homelab

Maybe consider not standing out...



Be the one to stand
out in the crowd.

Joel Osteen

 quoteafany

Synology/QNAP case study

- QNAP/Synology NAS exposed to public internet
- DeadBolt Ransomware Campaign exploited unpatched application vulnerabilities (i.e., QNAP's Photo Station)
 - Data/backup directories encrypted
 - Admin page hijacked with hard-coded Bitcoin ransom note



Synology/QNAP case study

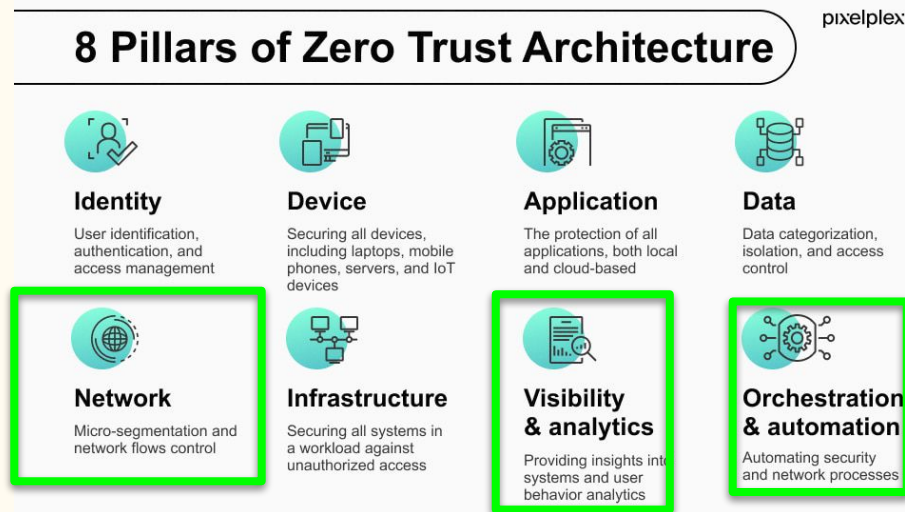
- Is it enough to remove public Internet accessibility?
 - Half the battle won
 - Compromising convenience
 - Can we prevent users from uploading executables? Can we prevent users from executing executables from NAS? Short answer: yes. Long answer: very difficult work, high effort required.
- Patching applications that can access Internet are also important
 - [How fast Windows XP gets compromised](#)
 - Avoid using EOL services
- VLAN is important, but also principle of infra security...



Principles of Infrastructure Security

Some foundations...

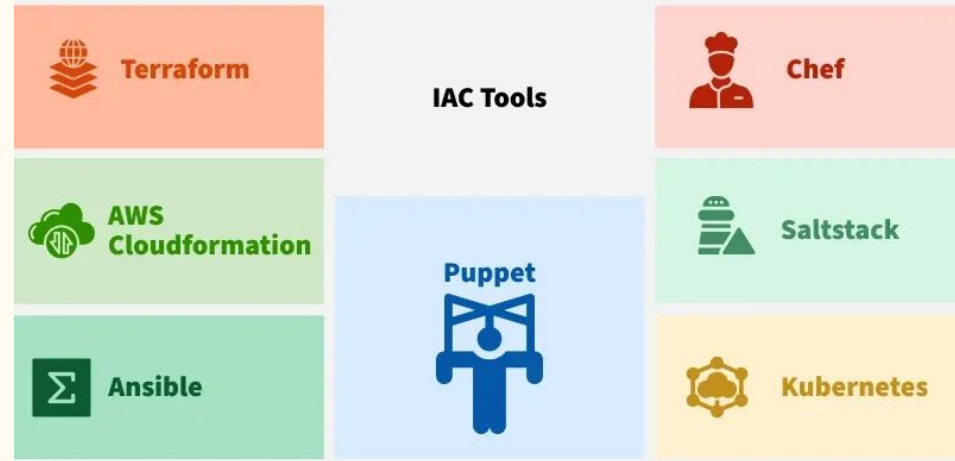
1. CIAAN (sure but its old-school)
2. **Foundation principles:**
 - a. Principle of Least Privilege, and Separation of Duties
 - b. Defense-in-Depth
 - i. Using Reverse Proxy instead of direct service
 - c. Security-by-Design
 - i. IaC/VLANs
 - ii. Continuous monitoring
 - d. Zero Trust (everything on the right)



<https://pixelplex.io/blog/how-to-implement-zero-trust/>

Infra as Code (IaC)

- TL;DR codified way to spin up your infra
- Terraform is the gold standard (or at least that is what I only know)
- Ansible/Chef/Puppet/Saltstack are known as a Configuration Management (CM) tool
- Kubernetes is like docker swarm, but on steroids. People often don't want to host kubernetes in homelab because it is overkill. But don't care what others say; it is your homelab.

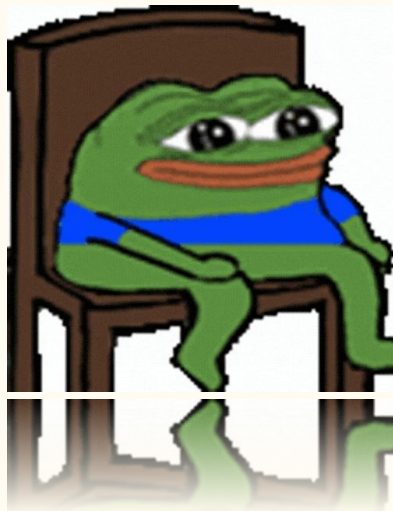


<https://www.geeksforgeeks.org/devops/what-is-infrastructure-as-code-iac/>

Terraform

- I will share a script today for you to bring back and try.
- Read the Terraform documentation, get started
- Understand the code I share with you
- Try running it on your VM
- See how it make things simpler
- Btw, the code is intentionally left incomplete; some fixing is required to force you to pick up Terraform.

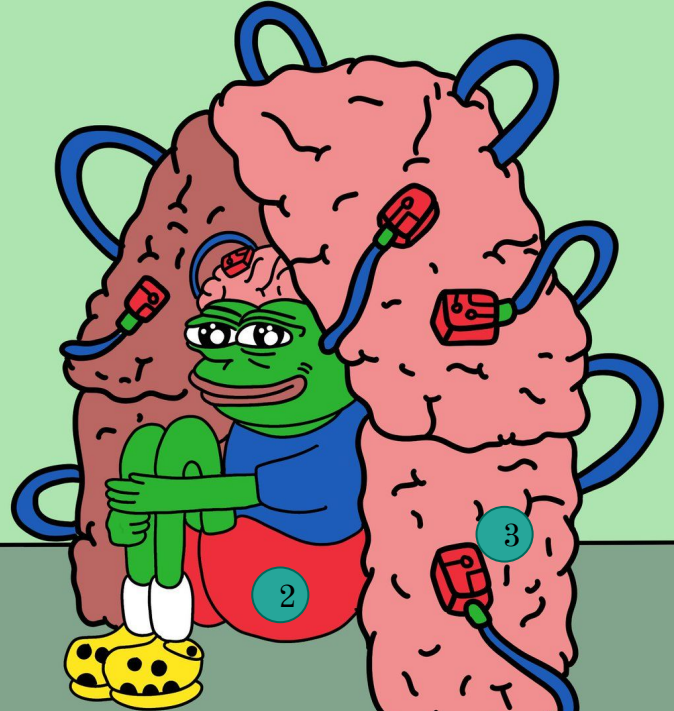
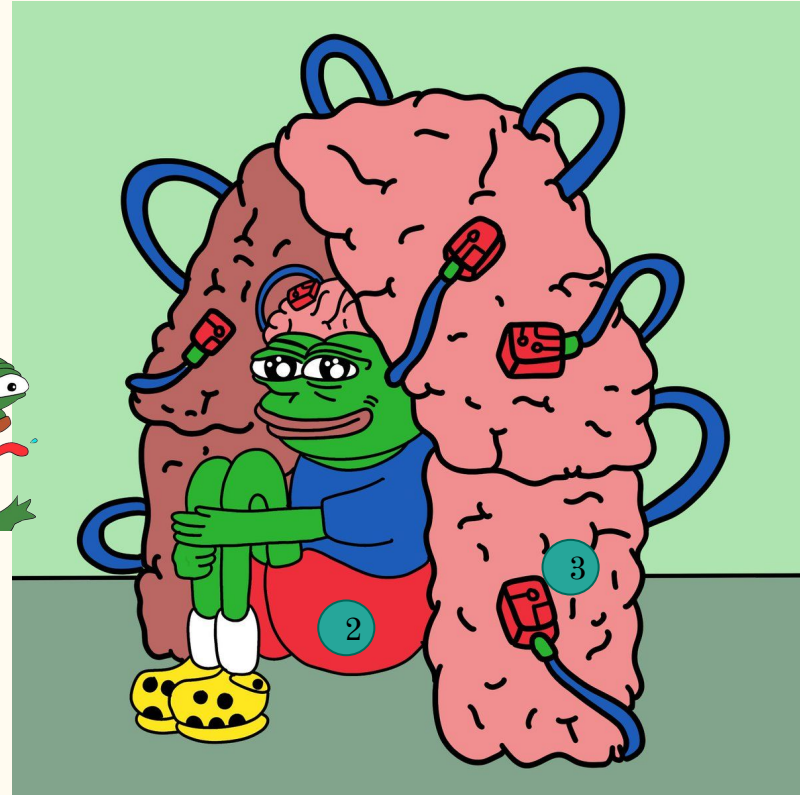
QUICK DEMO



MCP Servers

- What is MCP?
 - Model Context Protocol
 - Open source standard created by Anthropic
- Host, Client, Server. Which one on the right is host, client, server?
- Proxmox MCP

<https://github.com/RekklesNA/ProxmoxMCP-Plus>



Resources

1. [Homelab Almanac](#) (Full (paid) guide for self-hosting malware sandbox, DevSecOps, IaC)
2. [r/selfhosted](#)
 - a. https://www.reddit.com/r/selfhosted/comments/1u3cvwq/new_project_megathread_week_of_11_jun_2026/
3. [r/homelab](#)
4. [r/HomeLabPorn](#)
5. <https://www.skool.com/homelabexplorers>
6. <https://selfh.st>



Additional Docker tips



1. Auto Docker update images/containers

- nicholas-fedor/watchtower
- What's up Docker
- Diun
- Tugtainer

2. Other Docker management

(Dockge was chosen because it uses very little resources)

- Portainer: freemium model
- Dockhand
- Arcane
- Komodo

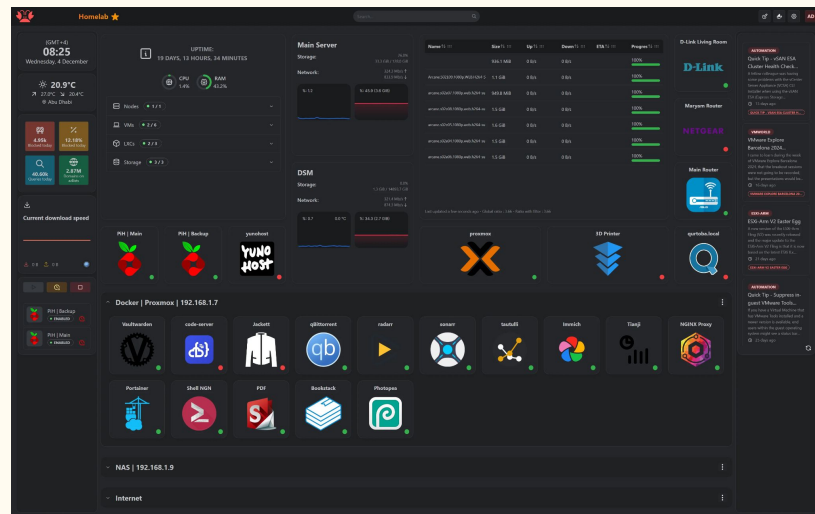
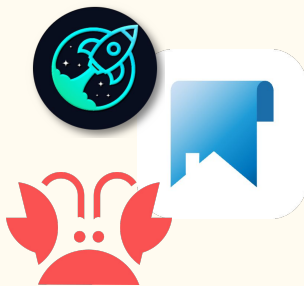
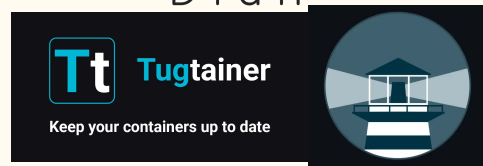
3. VPN bridge forward proxy

(I use AirVPN)

- Glutun

4. Homepage management

- Homepage
- Homarr
- Dashy
- Homer
- Flame



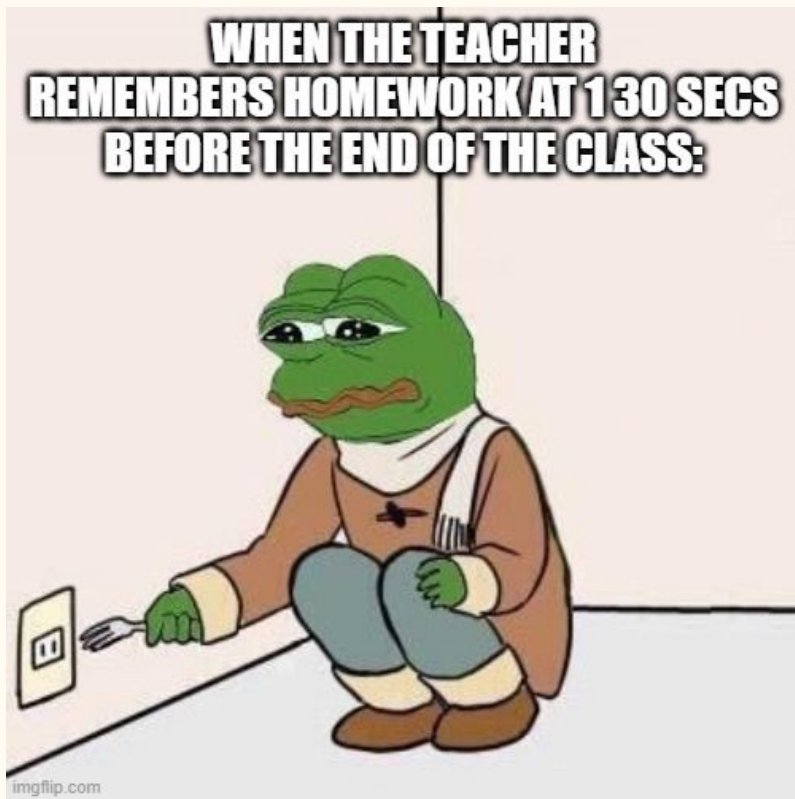
Closing remarks

- Cyber Security is a vast industry
- Intersect of many faculties:
 - Network security
 - DevSecOps
 - Infra security
- Development work is not necessary, but very useful
 - AI can aid this
 - MCP/Agentic
- Don't be afraid to break your homelab; everything is fixable (unless its hardware damage)
- Countless debate about purchasing 'just-enough' vs 'future-proofing' equipment
 - I initially bought just-enough, and then slowly stopped fulfilling my needs
 - Most importantly, start with what you have!



Homework (not really)

1. Install Proxmox on bare-metal
2. Deploy the service that will benefit you immediately (i.e., DNS)
3. Install MCP on Proxmox
 - a. Easier with Claude Code (not free)
4. Join the subreddits and div0 discord server
5. IaC your environment into Proxmox
 - a. Bonus: manage secrets using Hashicorp Vault
 - b. Use Jump host/box to set up the environment



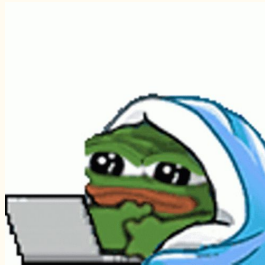
Credits

1. **Rowena** - thanks for organising and for the invite. Wouldn't have done this without you
2. **Anna** - for coordinating the admin!
3. **Kia Meng** - for the tremendous assist with the workshop since the start
4. **Emil** - for div0 and your guidance
5. **Dave** - for supplementing your technical expertise, time and effort for doing this at such a last minute request!!



Thanks and Bye

Jeremy, Jun Hong



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